UNITED STATES | ECONOMIC INDICATORS

US Macro Indicators: A Practitioner's Field Guide

What each release measures, how markets and the Fed react to it, and where to get the data free

PRIMER: READING MACRO DATA LIKE A STRATEGIST

Levels, changes and surprises

Markets price expectations, so the information content of a release is mostly the **surprise**: actual minus consensus (the median economist forecast compiled by Bloomberg, Reuters or similar). A 'strong' number that merely matches consensus is usually a non-event; a modestly weak number that undershoots a high consensus can move markets sharply. Revisions to prior months often matter as much as the headline - payrolls and GDP are the classic examples.

Diffusion indices (PMIs, sentiment)

Survey indicators such as the ISM and S&P Global PMIs are **diffusion indices**. Respondents say whether a variable is higher, the same or lower than the prior month. The index is % reporting higher plus half of % reporting the same (equivalently 50 + (%higher - %lower)/2). A reading of 50 separates expansion from contraction. A diffusion index measures the **breadth** of change, not its magnitude: 55 means more firms are improving than deteriorating, not that output rose 5%. Direction and momentum (rising vs falling) often carry more signal than the level.

Hard data vs soft data

Soft data (surveys: Michigan sentiment, PMIs) arrive first and capture expectations and turning points, but can be distorted by mood, politics and headlines. **Hard data** (payrolls, CPI, GDP) measure actual activity but arrive later and get revised. A persistent gap between soft and hard data (as in 2022-23, when surveys signalled recession and hard data did not) is itself a strategic signal: someone is going to be wrong.

Annualising, y/y vs m/m, seasonal adjustment

- **m/m** changes are timely but noisy; **y/y** changes are smooth but lag turning points and suffer **base effects** (last year's spike dropping out mechanically lowers y/y).
- **3- and 6-month annualised** rates ((1+m/m avg)^12 - 1) are the strategist's compromise; the Fed watches them closely for inflation.
- US GDP growth is quoted as a **seasonally adjusted annual rate (SAAR)** of the q/q change - roughly 4x the raw quarterly change. Most other countries quote raw q/q; do not compare them directly.
- Seasonal factors are re-estimated annually and can shift the monthly pattern ('residual seasonality', e.g. hot January inflation prints).

How to use this guide

Each chapter follows the same structure: **Bottom line**, **At a glance** (indicator type, lead/lag, key question, publisher, timing and Taipei release time), then **(1) Methodology**, **(2) What it implies & how it affects macro strategy** and **(3) How to get the data free**. Chapters are grouped by category: growth, surveys, labour, inflation, monetary policy, markets, energy and international institutions. **Release times:** most US data are released at 8:30 a.m. ET (21:30 Taipei); ISM, JOLTS and Michigan at 10:00 a.m. ET (23:00 Taipei). Subtract one hour while US daylight saving time is in force (mid-March to early November). FRED series codes are written in **BOLD CAPS**; paste them into fred.stlouisfed.org/series/CODE.

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CHAPTER 1 | GROWTH & OUTPUT

GDP / Real GDP

Bottom line

- The broadest but most **lagging** release: trade the surprise vs GDPNow and the composition.
- Real final sales to private domestic purchasers is the cleanest read on underlying demand.
- US recessions are dated by the NBER, not by the 'two negative quarters' rule.

At a glance

Indicator type	Hard data (activity)
Lead / lag	Lagging
Key question	Size and growth of the economy, what is driving it, and the output gap
Publisher	US Bureau of Economic Analysis (BEA)
Frequency	Quarterly: **Advance** estimate ~4 weeks after quarter end; **Second** and **Third** estimates follow monthly; annual updates (usually September) and periodic comprehensive revisions
Headline	Real GDP growth, q/q seasonally adjusted at annual rate (SAAR); also y/y and nominal GDP
FRED codes	GDP (nominal, $bn SAAR), GDPC1 (real, chained dollars), A191RL1Q225SBEA (real growth % SAAR), GDPDEF (deflator), GDPPOT (CBO potential)
Taipei time	21:30 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

Gross Domestic Product is the market value of all final goods and services produced within the US. BEA measures it primarily via the **expenditure approach**:

Exhibit 1: GDP by expenditure component

Component	Approx. share	Notes
C - Personal consumption	~68%	Goods (durable, non-durable) and services
I - Gross private domestic investment	~17-18%	Business fixed investment (equipment, structures, intellectual property incl. software and AI data centres), residential investment, and **change in private inventories** (very volatile)
G - Government consumption & investment	~17%	Federal (defence, non-defence) and state/local; excludes transfers like Social Security
NX - Net exports	~-3%	Exports minus imports; imports subtract because they are counted in C/I

Source: BEA; shares approximate.

Nominal vs real; chain weighting

Nominal GDP is measured in current dollars. **Real GDP** removes price changes using chain-type Fisher quantity indexes; levels are expressed in 'chained (2017) dollars'. Because of chaining, real components do not add exactly to the total. The **GDP price deflator** = nominal / real, a broad inflation measure that includes investment and government prices.

Alternative and complementary measures

- **GDI** (Gross Domestic Income) measures the same thing from the income side; the gap is the 'statistical discrepancy'. The average of GDP and GDI is often a better growth estimate.
- **Real final sales to private domestic purchasers** (C + private fixed investment) strips out volatile inventories, government and trade - the best 'core GDP' signal of underlying private demand. BEA now highlights it in releases.
- **Contributions to growth** (percentage points) tell you which component drove the headline.
- **Output gap:** real GDP vs CBO potential GDP (GDPPOT) indicates slack or overheating.

Revisions and distortions

The advance estimate relies on incomplete source data (BEA fills gaps with assumptions); revisions of 0.5-1pp in annualised growth are common, and annual revisions can reshape the past three years. Large swings in net exports and inventories around tariff deadlines (e.g. import front-loading in early 2025 produced a negative headline print while domestic demand was solid) are a reminder to look beneath the headline. Gold imports have also distorted trade data in recent years.

2. What It Implies & How It Affects Macro Strategy

Economic signal

GDP is the broadest measure of the economy but the most **lagging** of the major releases - by the advance estimate, markets have usually seen three months of payrolls, PMIs, retail sales and nowcasts. The market reaction depends on the surprise versus **nowcasts** (Atlanta Fed GDPNow, NY Fed Staff Nowcast) and on the composition (consumer and business investment strength vs inventory or trade noise).

How strategists use it

- **Trend vs potential:** potential growth is roughly 1.8-2% for the US (labour force growth + productivity); lower immigration reduces this. Growth persistently above potential with a closed output gap is inflationary; below potential raises unemployment (Okun's law: ~2pp of growth shortfall for a year ~ +1pp unemployment).
- **Recessions:** the US does not use the 'two negative quarters' rule. The NBER Business Cycle Dating Committee looks at depth, diffusion and duration across payroll employment, real income, real spending and industrial production. Two negative quarters in H1 2022 were not declared a recession.
- **Nominal GDP and earnings:** nominal GDP growth is the macro anchor for corporate revenue growth and for where long-term bond yields 'should' trade; 10-year yields well below nominal growth imply easy financial conditions.
- **Investment cycle:** business investment in equipment and intellectual property (software, AI, data centres) has been a major driver; strategists track it alongside capex guidance and semiconductor indices.
- **Productivity:** real GDP per hour (BLS productivity release) drives unit labour costs and non-inflationary growth capacity.

3. How to Get the Data Free

Exhibit 2: Where to get GDP data for free

Source	What you get	Notes
bea.gov/data/gdp	Releases, technical notes, full NIPA tables (Table 1.1.1 growth, 1.1.2 contributions, 1.1.6 real levels)	Interactive Data Application and release schedule
FRED / ALFRED	GDPC1, GDP, A191RL1Q225SBEA, GDI, GDPPOT; **ALFRED** keeps every vintage so you can see what was known at the time	Essential for backtesting
Atlanta Fed GDPNow	Running nowcast of current-quarter real GDP, updated after major releases	atlantafed.org/cqer/research/gdpnow
NY Fed Staff Nowcast	Weekly nowcast with news decomposition	newyorkfed.org
CBO	Potential GDP, output gap, long-term projections	cbo.gov/data
NBER	Official recession dates	nber.org/research/business-cycle-dating

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 2 | GROWTH & OUTPUT

Industrial Production & Capacity Utilisation

Bottom line

- Measures physical output of manufacturing, mining and utilities; **capacity utilisation** gauges slack.
- Confirms (or contradicts) what the ISM survey said two weeks earlier.
- Utilisation persistently above ~80% has historically coincided with building inflation pressure.

At a glance

Indicator type	Hard data (activity)
Lead / lag	Coincident
Key question	Momentum in factories, mines and utilities; spare capacity
Publisher	Federal Reserve Board, G.17 release 'Industrial Production and Capacity Utilization'
Frequency	Monthly, around the 15th-17th of the following month, 9:15 ET; annual revisions
Headline	Industrial Production Index (2017 = 100), m/m % change; total capacity utilisation rate (%)
FRED codes	INDPRO (total IP), IPMAN (manufacturing), TCU (total utilisation), MCUMFN (manufacturing utilisation)
Taipei time	22:15 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

Industrial production (IP) measures the real output of US factories, mines and utilities. The Fed builds the index from roughly 300 individual series, combined with **value-added weights** so that each industry counts in proportion to its contribution to the economy. Source data come in three forms: physical product counts (tons of steel, barrels of oil, motor-vehicle assemblies), **production-worker hours** from the BLS payroll survey, and **electricity use** by industry. Where only hours or power data exist at first, estimates are revised as better physical data arrive.

Exhibit 3: Industrial production by market and industry group

Group	Approx. share of IP	What drives it
Manufacturing	~75%	Durable goods (vehicles, machinery, computers, aerospace) and non-durables (food, chemicals, petroleum products); the cyclical core
Mining	~14%	Oil and gas extraction, drilling and coal; driven by energy prices and shale activity
Utilities	~11%	Electricity and gas; swings with weather (hot summers, cold winters) and, increasingly, data-centre power demand
Market groups	-	IP is also split into consumer goods, business equipment, construction supplies and materials - business equipment tracks capex

Source: Federal Reserve Board; shares approximate.

Capacity utilisation

Capacity utilisation = actual output / sustainable maximum output, where capacity is estimated from Census surveys of plant capacity and industry data. Total utilisation has averaged roughly 79-80% since 1972. Readings well above that level signal bottlenecks and pricing power; low readings (below ~75%) signal slack and weak capex incentives.

Quirks

- **Weather and autos** dominate monthly noise: utilities output swings with temperatures, and motor-vehicle assemblies with model changeovers and strikes (e.g. the 2023 UAW strike).
- IP covers only goods-producing industries - roughly a fifth of the economy - so it can contract while services drive overall growth.
- Annual revisions can change the profile of the past several years; production-worker hours make IP partly dependent on payroll data.

2. What It Implies & How It Affects Macro Strategy

Economic signal

IP is one of the four monthly series the NBER emphasises when dating recessions (with payroll employment, real income and real spending), and it peaked before or near most post-war recessions. In market terms it is a **confirmation** indicator: the ISM Manufacturing survey arrives two weeks earlier, so IP tells you whether the survey's message is showing up in actual output.

How strategists use it

- **ISM vs IP:** a rising ISM with flat IP suggests sentiment ahead of reality (or an inventory build); persistently weak IP with ISM above 50 casts doubt on the survey.
- **Inflation pressure:** utilisation above ~80-82% has historically coincided with rising goods inflation and PPI; low utilisation dampens pricing power.
- **Capex and reshoring:** business-equipment output, high-tech output and utilisation in semiconductors and electrical equipment show whether industrial policy and AI investment are adding capacity.
- **Energy:** mining output reflects US oil and gas production trends and links to the oil chapter.
- **Cyclical equities:** materials, industrials and transport stocks correlate with IP momentum; Taiwan and Korean exporters watch US goods demand closely.

3. How to Get the Data Free

Exhibit 4: Where to get industrial production data for free

Source	What you get	Notes
federalreserve.gov/releases/g17	Release, full tables by industry and market group, capacity and utilisation, historical data	Data Download Program (DDP) gives CSV of every series
FRED	INDPRO, IPMAN, IPB50001N, TCU, MCUMFN and hundreds of industry sub-indices	Easy charting and API
Regional Fed surveys	Philadelphia, Empire State, Richmond, Kansas City, Dallas manufacturing surveys	Earlier monthly reads on factory activity

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 3 | GROWTH & OUTPUT

Housing: Starts & Permits (新屋開工 / 營建許可), New Home Sales (新屋銷售), Existing Home Sales (成屋銷售)

Bottom line

- **Permits lead starts**, and housing leads the business cycle: it is the first sector to feel rate changes.
- New home sales are counted at contract and are timely but noisy; existing sales (about 90% of transactions) are counted at closing.
- Mortgage rates track the 10-year Treasury, so housing data feed straight back into the rates view.

At a glance

Indicator type	Hard data (activity)
Lead / lag	Leading (permits, starts) / coincident (sales)
Key question	The most rate-sensitive sector: is housing turning the cycle?
Publishers	**Starts & permits**: Census Bureau & HUD, 'New Residential Construction'. **New home sales**: Census & HUD, 'New Residential Sales'. **Existing home sales**: National Association of Realtors (NAR)
Frequency	Monthly. Starts/permits ~12th business day (8:30 ET); new home sales ~17th business day (10:00 ET); existing home sales ~3rd-4th week (10:00 ET)
Headline	Seasonally adjusted annual rates (SAAR) in thousands (starts, permits, new sales) or millions (existing sales)
FRED codes	HOUST, HOUST1F (single-family starts), PERMIT, PERMIT1, HSN1F (new home sales), MSACSR (months' supply), MSPNHSUS (median new home price), CSUSHPINSA (Case-Shiller), MORTGAGE30US
Taipei time	Starts & permits 21:30; new and existing home sales 23:00 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

Exhibit 5: The three housing activity releases compared

Release	What is counted	When it is counted	Key quirks
Building permits	New housing units authorised by ~20,000 permit-issuing places	When the permit is issued	Least noisy of the three; leads starts by about 1-2 months; part of the Conference Board LEI
Housing starts	Units on which excavation begins (Survey of Construction)	Start of construction	Multifamily (5+ units) is lumpy; also reports units under construction and completions
New home sales	Newly built single-family homes sold	At **contract signing** (deposit)	Timely but noisy: monthly 90% confidence interval often +/-15-20%; large revisions; includes months' supply and median price
Existing home sales	Resales of existing single-family homes, condos and co-ops (from local MLS data)	At **closing** (1-2 months after contract)	About 90% of transactions; NAR's Pending Home Sales Index (contracts) leads it; median price is not quality-adjusted

Source: Census Bureau, HUD, NAR.

Other key housing measures

- **Months' supply** = inventory / sales pace. About 6 months is traditionally a balanced market; above it favours buyers, below it sellers.
- **House prices:** median prices in the sales reports shift with the mix of homes sold. Use repeat-sales indices - S&P CoreLogic Case-Shiller and FHFA - for true price trends.
- **Builder sentiment:** the NAHB/Wells Fargo Housing Market Index (mid-month) is a timely soft read on new-home demand.
- **Mortgage rates:** Freddie Mac's weekly 30-year fixed rate (FRED: MORTGAGE30US) usually trades ~1.5-3pp over the 10-year Treasury yield.

2. What It Implies & How It Affects Macro Strategy

'Housing is the business cycle'

Residential investment is only ~3-4% of GDP but it is the most **interest-rate-sensitive** sector, and it has turned down ahead of most post-war recessions (Edward Leamer's well-known argument). Building permits and starts react within months to changes in mortgage rates; housing then feeds employment (construction, real estate), durable-goods demand (furniture, appliances) and household wealth.

Recent structural features

- **Rate lock-in:** most US mortgages are 30-year fixed; owners who refinanced at 3% in 2020-21 are reluctant to sell, depressing existing-home inventory and sales while supporting prices.
- **Builders gained share:** with resale inventory scarce, builders captured more demand, using mortgage-rate buydowns to sustain sales - so new home sales held up better than existing sales.
- **Shelter inflation link:** rents and house prices feed CPI shelter with a lag, so housing data inform the inflation outlook as well as growth.

How strategists use it

- **Rates transmission:** falling permits after a rise in mortgage rates confirms that policy is biting; a pickup in permits after rate cuts is one of the first signs of an easing cycle working.
- **Cycle timing:** single-family permits rolling over while payrolls are still strong is a classic late-cycle warning.
- **Equities:** homebuilders (ITB/XHB ETFs), building products, mortgage lenders and home-improvement retailers trade off these releases.
- **Watch the right series:** trade single-family permits and pending home sales for direction; ignore one-month swings in multifamily starts and new home sales.

3. How to Get the Data Free

Exhibit 6: Where to get housing data for free

Source	What you get	Notes
census.gov/construction	New Residential Construction and New Residential Sales releases, full history, regional detail	Time-series tool exports CSV
FRED	HOUST, PERMIT, HSN1F, MSACSR, CSUSHPINSA, MORTGAGE30US, USSTHPI (FHFA)	Free API
nar.realtor	Existing home sales and pending home sales headlines, median prices, months' supply	Detailed history is limited; FRED shows only recent months of NAR data
nahb.org	NAHB/Wells Fargo Housing Market Index	Free history
freddiemac.com/pmms	Weekly mortgage rate survey	Since 1971

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 4 | GROWTH & OUTPUT

Current Account / GDP and Budget Balance / GDP

Bottom line

- The US runs persistent **current account** and **budget** deficits - the 'twin deficits'.
- Deficits are financed by foreign capital inflows and Treasury issuance, linking them to the dollar and term premium.
- Fiscal deficits near 6% of GDP outside recession are historically unusual and keep long-end yields sensitive.

At a glance

Indicator type	Hard data (external & fiscal)
Lead / lag	Slow-moving / structural
Key question	How the US finances itself abroad, and how much Treasury debt the market must absorb
Publishers	**Current account**: BEA, U.S. International Transactions (quarterly); monthly trade balance from BEA & Census. **Budget balance**: US Treasury, Monthly Treasury Statement; CBO projections
Frequency	Trade balance monthly (~5 weeks after month-end, 8:30 ET); current account quarterly (~11-12 weeks after quarter-end); Monthly Treasury Statement ~8th business day (2:00 p.m. ET); CBO Monthly Budget Review a few days after month-end
Headline	Current account balance ($bn and % of GDP); federal surplus/deficit ($bn, fiscal year to date, % of GDP)
FRED codes	IEABC (current account), BOPGSTB (trade balance), MTSDS133FMS (monthly federal surplus/deficit), FYFSGDA188S (federal surplus/deficit % of GDP), GFDEGDQ188S (federal debt % of GDP)
Taipei time	Trade & current account 21:30; Monthly Treasury Statement 03:00 next day (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

Current account

Exhibit 7: Balance of payments structure

Account	Contents	US position
Goods & services trade	Exports minus imports	Large goods deficit, partly offset by a services surplus (travel, financial and IP services)
Primary income	Investment income (interest, dividends) and compensation	Historically positive, but eroding as US external liabilities and interest rates rise
Secondary income	Transfers (remittances, aid)	Small, persistent deficit
= Current account	Sum of the above	Persistent deficit, roughly 3-4% of GDP in recent years; it widened sharply in early 2025 as importers front-ran tariffs
Financial account	Net purchases of foreign vs US assets	Mirror image: foreigners' net purchases of US Treasuries, equities, corporate bonds and FDI finance the deficit

Source: BEA.

Accounting identity: the current account equals national saving minus domestic investment (CA = S - I). A deficit means the US invests more than it saves and imports capital. The cumulative result is a deeply negative **net international investment position** (US liabilities to foreigners exceed US assets abroad by tens of trillions of dollars).

Budget balance

The **Monthly Treasury Statement (MTS)** reports federal receipts, outlays and the deficit on a cash basis for the fiscal year (October-September). Monthly figures are distorted by calendar shifts (benefit payments moved when the 1st falls on a weekend) and tax dates (April, June, September, December), so compare fiscal year-to-date totals with the prior year or use 12-month rolling sums. The **Daily Treasury Statement** gives near-real-time cash flows (including tariff revenue). The **Congressional Budget Office (CBO)** publishes the Budget and Economic Outlook (early in the year), long-term projections and cost estimates of legislation.

- Federal deficits of roughly 6% of GDP in FY2024-25 - with unemployment near 4% and no recession - are historically unusual outside wartime.
- Net interest costs now exceed defence spending, so higher rates feed directly into larger deficits.
- Customs duties surged in 2025 as new tariffs took effect, one of the few large offsets on the revenue side.

2. What It Implies & How It Affects Macro Strategy

Why these slow-moving balances matter for markets

- **Treasury supply and term premium:** deficits must be financed by issuance. Treasury's **Quarterly Refunding Announcement** (bill vs coupon mix) has become a market event; heavier long-dated supply raises the term premium and steepens the curve.
- **Fiscal dominance risk:** large deficits with high debt (~100% of GDP held by the public) raise questions about debt sustainability, credit ratings (Moody's removed the last AAA rating in May 2025) and pressure on the Fed.
- **Fiscal impulse:** the change in the deficit, not its level, drives growth; a shrinking deficit is a drag.
- **Twin deficits and the dollar:** a current account deficit requires capital inflows. If foreign appetite for US assets fades (e.g. reserve diversification, hedging by Asian investors), the dollar and Treasuries must adjust.
- **Trade policy:** monthly trade data show tariff front-running and diversion (e.g. imports shifting from China to Vietnam, Mexico and Taiwan) and swing quarterly GDP via net exports.
- **Taiwan angle:** Taiwan's large bilateral surplus with the US (driven by semiconductors and AI servers) makes it sensitive to US trade policy and to the dollar.

3. How to Get the Data Free

Exhibit 8: Where to get external and fiscal data for free

Source	What you get	Notes
bea.gov (International Transactions; Trade in Goods & Services)	Quarterly current account, NIIP, monthly trade by country and product	Interactive data tables
census.gov/foreign-trade	Monthly goods trade by country and commodity; advance goods trade balance	USA Trade Online free
fiscaldata.treasury.gov	Monthly and Daily Treasury Statements, debt to the penny, interest costs, auction data	API, CSV
home.treasury.gov	Quarterly Refunding statements and TBAC materials	Market-moving issuance guidance
cbo.gov/data	Budget projections, long-term outlook, historical budget data	Excel files
FRED; IMF data portal	IEABC, BOPGSTB, FYFSGDA188S, GFDEGDQ188S; IMF for cross-country % of GDP comparisons	Free

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 5 | BUSINESS & CONSUMER SURVEYS

ISM PMI: Manufacturing & Services (and Sub-indices)

Bottom line

- ISM Manufacturing (since 1948) is the market's favourite **business-cycle** gauge and correlates with earnings, yields, copper and credit.
- Prices Paid leads goods inflation; Employment previews payrolls; New Orders minus Inventories leads production.
- Manufacturing is ~10% of GDP: always cross-check with ISM Services.

At a glance

Indicator type	Soft data (business survey)
Lead / lag	Leading
Key question	Industrial cycle, the prices pipeline and recession risk
Publisher	Institute for Supply Management (ISM), Report On Business (ROB)
Frequency	Monthly. **Manufacturing**: 1st business day, 10:00 ET. **Services**: 3rd business day, 10:00 ET
History	Manufacturing since 1948 - one of the longest-running US cyclical indicators. Services since 1997 (renamed from 'Non-Manufacturing' to 'Services PMI' in 2020)
Panel	Purchasing and supply executives across ~18 industries in each sector, weighted by GDP contribution
Taipei time	23:00 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

ISM members answer whether each business variable is better/higher, the same, or worse/lower than the prior month. Each index = % positive + 0.5 x % same, seasonally adjusted where seasonality exists. ISM also reports, for each index, the percentage of respondents answering higher/same/lower, and a list of industries reporting growth or contraction ranked by strength.

Manufacturing PMI: equal-weighted (20% each)

Exhibit 9: ISM Manufacturing PMI: five equal-weighted components

Component	What it measures	Notes
New Orders	Incoming orders	Most closely watched leading component
Production	Output levels	Tracks Fed industrial production
Employment	Headcount trends	Leads manufacturing payrolls
Supplier Deliveries	Speed of vendor deliveries	Index above 50 = **slower** deliveries (counted as positive because it usually signals strong demand). Not inverted in the published sub-index
Inventories	Manufacturers' own inventories	Rising = building stock

Source: Institute for Supply Management.

Services PMI: equal-weighted (25% each)

Business Activity, New Orders, Employment and Supplier Deliveries. Unlike S&P Global's single-question services headline, ISM Services is a composite.

Other sub-indices (細項) - not in the headline but heavily watched

Exhibit 10: ISM sub-indices outside the headline

Sub-index	Mfg	Svcs	Signal
Prices	Yes	Yes	Share of firms paying higher prices. Mfg Prices leads PPI/goods CPI; Services Prices is a leading gauge of services inflation. Note: not seasonally adjusted in some periods; volatile
Backlog of Orders	Yes	Yes	Unfilled orders; capacity strain
New Export Orders	Yes	Yes	Foreign demand
Imports	Yes	Yes	Input sourcing from abroad; tariff front-running shows here
Customers' Inventories	Yes	-	Below 50 = customers' stocks 'too low' (bullish for future orders). Combined with New Orders it is a strong leading signal
Inventory Sentiment	-	Yes	Are inventories too high or too low?

Source: Institute for Supply Management.

ISM's own breakeven levels

ISM publishes estimates linking its indices to the broader economy: a Manufacturing PMI above roughly the low-to-mid 40s (ISM's long-run estimate has been around 42-43) has historically been consistent with expansion of **overall** GDP, even though 50 is the line for manufacturing itself. ISM also publishes thresholds for the Employment, New Orders and Services indices relating them to BLS employment and Census data. Check each report's footnotes for current values.

2. What It Implies & How It Affects Macro Strategy

Why ISM Manufacturing is the market's favourite cycle indicator

- **Long history and timeliness:** available on the first business day, before payrolls, with data back to 1948 covering every post-war recession.
- **Asset correlations:** the ISM Manufacturing PMI correlates strongly with y/y S&P 500 earnings growth, cyclical-vs-defensive equity performance, copper, the 10-year Treasury yield, credit spreads and EM equities. Many macro desks frame the business cycle as 'ISM rising/falling, above/below 50'.
- **Recession signal:** sustained readings in the low 40s have coincided with recessions; but manufacturing is only ~10% of US GDP, so the 2022-24 manufacturing recession did not translate into an economy-wide downturn. Always cross-check with Services.
- **Inflation pipeline:** the Prices Paid index leads goods inflation by several months and is the first monthly hard number on tariff and commodity pass-through.

High-signal combinations

Exhibit 11: High-signal ISM combinations

Combination	Reading	Typical implication
New Orders - Inventories spread	Positive and rising	Production will be ramped up; early-cycle, pro-cyclical
New Orders < 50 and Customers' Inventories > 50	Demand weak, customers overstocked	Destocking phase; avoid cyclicals, favour bonds
Prices > 60 with New Orders < 50	Stagflationary mix	Hard for the Fed; bear-flattening risk, weak equities
Services Employment < 50, Mfg Employment < 50	Broad labour weakening	Raises odds of weak payrolls; front-end rallies
Supplier Deliveries spike with weak orders	Supply shock, not demand	Headline PMI overstates strength; look through it

Source: compiled for this guide.

Trading the release

Rates and FX react to the **headline surprise and the Employment/Prices components** (Employment as a payroll preview; Prices for inflation). Equities react more to New Orders. Read the 'What Respondents Are Saying' quotes - they frequently anticipate policy-driven shifts (tariffs, capex delays) before the numbers show them.

3. How to Get the Data Free

Exhibit 12: Where to get ISM data for free

Source	What you get	Notes
ismworld.org > Supply Management News and Reports > ROB	Full current-month Manufacturing and Services reports: all sub-indices, industry rankings, respondent comments	Free for the current release; deep history requires ISM membership/data subscription
Economic calendars and TradingEconomics	Headline and major sub-index histories with consensus	Good free workaround for history
Not on FRED	ISM asked FRED to remove its data in 2016	Free proxies on FRED: regional Fed surveys - Philadelphia (GACDFSA066MSFRBPHI), Empire State (GACDINA066MNFRBNY), Richmond, Kansas City, Dallas
Regional Fed websites	Full sub-index histories (prices, new orders, employment) for each district survey	Released 1-2 weeks before ISM: a good ISM preview

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 6 | BUSINESS & CONSUMER SURVEYS

S&P Global US PMI (and Sub-indices)

Bottom line

- The **flash PMI** is the earliest monthly read on US, Eurozone, UK, Japan and India activity.
- Same methodology in ~40 countries makes it the standard for **global cycle** comparisons.
- Watch Prices Charged in services for core inflation and New Orders vs Output for turning points.

At a glance

Indicator type	Soft data (business survey)
Lead / lag	Leading / coincident
Key question	Momentum in manufacturing and services; the flash is the earliest monthly read
Publisher	S&P Global Market Intelligence (formerly IHS Markit / Markit Economics)
Frequency	Monthly. **Flash** estimate ~3rd-4th week of the reference month (9:45 ET); **Final** Manufacturing on the 1st business day and Services/Composite on the 3rd business day of the next month
Headline series	US Manufacturing PMI, US Services Business Activity Index, US Composite PMI Output Index
Panel	Roughly 800 manufacturers and 800 service-sector companies, stratified by industry and firm size
Taipei time	22:45 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

Purchasing and senior executives answer a standardised questionnaire in the second half of each month, reporting whether each variable is higher, the same or lower than one month earlier. Each sub-index is a diffusion index (% higher + 0.5 x % unchanged) and is seasonally adjusted. The same methodology is used in roughly 40 countries, which makes S&P Global PMIs the standard for **cross-country comparison** (Eurozone, China Caixin/RatingDog, Japan, UK, India etc.).

Manufacturing PMI: weighted composite of five sub-indices

Exhibit 13: S&P Global Manufacturing PMI: component weights

Component	Weight	Interpretation
New Orders	30%	Demand pipeline; the most forward-looking component
Output	25%	Current production volumes
Employment	20%	Hiring vs shedding staff
Suppliers' Delivery Times	15% (inverted)	Longer delivery times usually signal strong demand, so the index is inverted (slower deliveries raise the PMI). Supply shocks (COVID, ports) can distort this upward
Stocks of Purchases	10%	Inventory of inputs; rising stocks = confidence/precautionary buying

Source: S&P Global Market Intelligence.

Services and Composite

The headline **Services Business Activity Index** comes from a single question: is the level of business activity higher, the same or lower than a month ago? The **Composite PMI Output Index** is a weighted average of the Manufacturing Output Index and the Services Business Activity Index, weighted by each sector's share of GDP (services dominate, so the composite tracks services closely).

Key sub-indices (細項) and what they tell you

Exhibit 14: S&P Global PMI sub-indices and what they signal

Sub-index	Sector	Signal
New Orders / New Business	Both	Leading demand; New Orders minus Inventories is a popular cycle indicator
New Export Orders	Both	External demand; sensitive to the dollar and tariffs
Backlogs of Work	Both	Capacity pressure; rising backlogs precede hiring and price increases
Input Prices (Input Costs)	Both	Cost pressure; leads PPI and goods CPI by ~2-6 months
Output Prices / Prices Charged	Both	Pricing power; the best survey proxy for pass-through to CPI/PCE
Employment	Both	Leads payroll trends at turning points
Future Output / Business Expectations	Both	12-month sentiment (not seasonally adjusted); tracks equity markets
Stocks of Finished Goods	Mfg	Unwanted inventory build = coming production cuts
Quantity of Purchases	Mfg	Purchasing activity; early read on industrial demand

Source: S&P Global Market Intelligence.

2. What It Implies & How It Affects Macro Strategy

Economic signal

The S&P Global PMIs are designed to track **GDP and industrial production growth** in near real time. The composite output index has historically correlated well with quarterly GDP growth, and S&P Global publishes rules of thumb mapping composite readings to GDP. Because the **flash** release comes ~1 week before the month ends, it is the **earliest hard-number read on a month** available for the US, the Eurozone, the UK, Japan, India and Australia, all published on the same day.

ISM vs S&P Global: why they diverge

- **Weights:** S&P Global weights New Orders 30% and inverts deliveries; ISM equal-weights its five components.
- **Panel:** ISM's panel is skewed to large, often multinational firms; S&P Global's panel is broader by firm size.
- **Seasonal adjustment and timing** differ, and services definitions differ (ISM's services headline is a composite; S&P's is a single activity question).
- In practice, markets react more to ISM, but S&P Global's flash moves markets on its day and is better for international comparison. When the two disagree, treat the signal as low conviction and look to sub-indices.

How strategists use it

- **Global cycle synchronisation:** comparing US, Eurozone, China and EM manufacturing PMIs shows whether the global industrial cycle is broadening or diverging - a key input for USD direction, EM equities and commodity exposure.
- **Inflation pipeline:** Input Prices and Prices Charged in services are watched as leading indicators of core services inflation, the Fed's stickiest category.
- **Recession timing:** a composite falling below 50 with New Orders weakening faster than Output, and Employment turning negative, is a classic late-cycle sequence.
- **Sector rotation:** rising manufacturing PMI favours cyclicals, materials, industrials and small caps; falling PMIs favour duration (Treasuries), quality and defensives.
- **Commentary is data:** the release text from the chief business economist often quantifies tariff, supply-chain or election effects, which helps attribute moves.

3. How to Get the Data Free

Exhibit 15: Where to get S&P Global PMI data for free

Source	What you get	Notes
pmi.spglobal.com	Free press releases (PDF) for every country on release day, with headline numbers, key sub-index commentary, charts and a release calendar	Full historical sub-index data requires a paid subscription; free registration gives release alerts
Economic calendars (Investing.com, ForexFactory, TradingEconomics)	Flash and final headline values, consensus and prior readings; downloadable short histories	TradingEconomics charts long history free on-screen; downloads may require an account
S&P Global commentary pages	Chief economist analysis and charts of sub-indices	Good for qualitative colour
Not on FRED	S&P Global PMIs are licensed data	Use regional Fed surveys on FRED as free proxies

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 7 | BUSINESS & CONSUMER SURVEYS

University of Michigan Consumer Sentiment Index

Bottom line

- Best read as a gauge of household **inflation psychology** first and a growth signal second.
- The 1-yr vs 5-10-yr inflation expectations gap separates transitory price shocks from de-anchoring risk.
- Partisan swings and the 2024 move to web surveys add noise: trade the 3-month trend, not one print.

At a glance

Indicator type	Soft data (household survey)
Lead / lag	Leading (weakly)
Key question	Consumer mood and household inflation expectations
Publisher	Surveys of Consumers, Institute for Social Research, University of Michigan (since 1946; index base Q1 1966 = 100)
Frequency	Monthly. Preliminary reading ~2nd Friday of the month; Final ~last/second-to-last Friday; 10:00 ET
Key outputs	Index of Consumer Sentiment (ICS), Current Economic Conditions (ICC), Index of Consumer Expectations (ICE), 1-yr and 5-10-yr inflation expectations
FRED codes	UMCSENT (sentiment), MICH (1-yr median inflation expectation)
Taipei time	23:00 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

The survey asks a nationally representative sample of US adults around 50 core questions about personal finances, business conditions, buying conditions and prices. The headline **Index of Consumer Sentiment** is built from just five questions:

Exhibit 16: The five questions behind the Index of Consumer Sentiment

#	Question (paraphrased)	Feeds
Q1	Are you (and your family) better or worse off financially than a year ago?	Current conditions
Q2	Will you be better or worse off financially a year from now?	Expectations
Q3	Will business conditions in the country be good or bad over the next 12 months?	Expectations
Q4	Will the country have good times or widespread unemployment/depression over the next 5 years?	Expectations
Q5	Is now a good or bad time to buy major household items (furniture, appliances, TVs)?	Current conditions

Source: University of Michigan Surveys of Consumers.

For each question a **relative score** = (% favourable - % unfavourable) + 100 is computed. The ICS equals the sum of the five relative scores divided by the 1966 base-period total (6.7558) plus a constant of 2.0 to correct for sample design changes in the 1950s. The **Current Conditions** sub-index uses Q1 and Q5; the **Expectations** sub-index uses Q2-Q4 (and historically fed into leading-indicator composites).

Inflation expectations

Respondents give the percentage they expect prices to change over the next year and over the next 5-10 years. The published figures are **medians** (means are skewed by extreme answers). These series are among the most cited household inflation-expectation gauges at the Fed, alongside the New York Fed's Survey of Consumer Expectations.

Important quirks

- **Mode change:** the survey moved from telephone interviews to web-based interviewing (fully by mid-2024). Web samples tend to produce lower sentiment and somewhat higher, more dispersed inflation expectations, so pre- and post-transition levels are not perfectly comparable.
- **Small sample, big noise:** the preliminary reading is based on a subset of interviews and is regularly revised at the Final release. Moves of 2-3 points are usually within sampling error.

- **Partisan bias:** since roughly 2016 the gap between Democrats' and Republicans' responses has widened sharply and flips after elections. A large share of 'sentiment shocks' around elections reflects politics rather than spending intentions.
- **Price-level sensitivity:** households react to the price *level* (and gasoline/food prices) not just the inflation rate, so sentiment can stay depressed long after inflation has normalised.
- The University of Michigan restricts some details (e.g. full preliminary tables) for a short embargo period; the headline numbers are public on release day.

2. What It Implies & How It Affects Macro Strategy

Economic signal

Sentiment is a **mood and expectations** gauge, not a spending forecast. Its correlation with real consumption growth is positive but loose: people often say they feel bad while continuing to spend (the 2022-24 'vibecession'). It is most informative at **extremes and turning points**: deep, broad-based collapses (2008, 2020) coincide with recessions, and recoveries in expectations have historically led consumer-spending recoveries. The Expectations sub-index is more forward-looking than Current Conditions.

Why the Fed and bond market care: inflation expectations

The more market-moving part of the release is often the **inflation expectations** line. The Fed's framework relies on expectations staying anchored near 2%. A jump in the **5-10 year** median (e.g. above its 2.9-3.1% historical range) raises the risk the Fed keeps policy tighter for longer; Chair Powell cited a preliminary Michigan expectations jump in justifying the 75bp hike of June 2022. Traders watch this line for front-end rate and breakeven-inflation (TIPS) reactions.

How strategists use it

- **Consumer-discretionary vs staples:** sustained improvement in expectations supports cyclical retail, autos and housing-related equities; deterioration favours defensives.
- **Cross-check with hard data:** compare with real personal spending, retail sales and the Conference Board Consumer Confidence Index (which weights the labour market more heavily, so it tracks payrolls better; Michigan tracks prices and personal finances more).
- **Read the sub-questions:** buying conditions for durables, homes and vehicles, and the share of respondents mentioning high prices or tariffs, provide colour on specific sectors.
- **Fade political noise:** when the partisan split explains most of a move, the market impact usually fades. Look at independents or at the cross-party average change.
- **Term structure of expectations:** 1-yr expectations spiking while 5-10-yr stay flat = transitory price shock (e.g. tariffs, gasoline); both rising = de-anchoring risk, which is a hawkish signal.

> **Rule of thumb**
> Treat Michigan as a high-frequency read on household inflation psychology first, and a growth indicator second. Never trade a single month's headline in isolation; trade the 3-month trend, the expectations lines and the surprise versus consensus.

3. How to Get the Data Free

Exhibit 17: Where to get Michigan sentiment data for free

Source	What you get	Notes
sca.isr.umich.edu	Headline ICS, ICC, ICE, inflation expectations; long historical tables (since 1952/1978); monthly press summary and charts	'Data' > 'Tables' lets you download full time series as CSV/Excel
FRED: UMCSENT, MICH	Monthly sentiment and 1-yr expected inflation; easy charting and API access	FRED may lag the preliminary print; use for history
NY Fed SCE (newyorkfed.org)	Alternative household inflation & labour expectations	Free microdata and charts
Conference Board	Consumer Confidence headline via press release	Full history is paid; headline is free

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 8 | LABOUR MARKET

BLS Employment Situation Report & Non-farm Payrolls Change

Bottom line

- The **Employment Situation** report is the single biggest monthly market mover.
- Use the 3-month average of payrolls with revisions and compare to a much lower post-2025 **breakeven** pace.
- Annual QCEW benchmarks can revise levels by hundreds of thousands of jobs.

At a glance

Indicator type	Hard data (labour)
Lead / lag	Coincident
Key question	Labour demand; the single biggest monthly market mover
Publisher	BLS, 'The Employment Situation' report
Frequency	Monthly, usually the first Friday of the month, 8:30 ET (covering the prior month)
Headline	Monthly change in total non-farm payroll employment (thousands, SA), plus prior two months' revisions
FRED codes	PAYEMS (level; use 'Change' transform), USPRIV (private), CES0500000003 (avg hourly earnings), AWHAETP (avg weekly hours)
Taipei time	21:30 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

The Employment Situation report

The BLS **Employment Situation** report, usually released on the first Friday of each month, combines two separate surveys. The **establishment survey** (Current Employment Statistics, CES) produces non-farm payrolls, average hourly earnings and average weekly hours by industry. The **household survey** (Current Population Survey, CPS) produces the unemployment rate, labour-force participation and demographic detail (see the next chapter). The two can diverge for months because they measure different things - jobs versus employed people - with different samples.

Non-farm payrolls

Payrolls come from the **Current Employment Statistics (CES)** or 'establishment survey' of roughly 120,000 businesses and government agencies covering about 630,000 individual worksites (roughly a quarter to a third of all US payroll jobs). Employers report the number of people on payroll for the pay period including the 12th of the month. A person with two jobs is counted twice; farm workers, the self-employed and private household workers are excluded.

Key mechanics

- **Revisions:** each month's first estimate is revised in each of the next two releases as more survey responses arrive. Sequences of large downward revisions are an early warning of weakening.
- **Annual benchmark:** CES is benchmarked to the near-census **Quarterly Census of Employment and Wages (QCEW)** from unemployment-insurance records. BLS publishes a preliminary benchmark estimate in late summer (the preliminary 2025 benchmark implied about 911,000 fewer jobs in March 2025 than first reported) and incorporates the final benchmark with the January report released in February.
- **Birth-death model:** BLS imputes jobs from business openings and closings not captured by the sample. It tends to overstate jobs at turning points into downturns.
- **Same release, other key series:** average hourly earnings (wage growth), average weekly hours, the diffusion index of industries adding jobs, and sector detail (health care, government, leisure, manufacturing, temp help - a classic leading sector).
- **Distortions:** strikes, hurricanes, federal workforce changes (e.g. deferred-resignation programmes in 2025), census hiring and calendar effects (4 vs 5 week gaps between surveys) can move the headline.

The breakeven pace

The 'breakeven' payroll gain is the number needed to keep the unemployment rate stable, determined by labour force growth. During 2022-24, strong immigration pushed estimates to 150-250k a month. With much lower net immigration and an ageing population since 2025, many estimates fell to well under 100k - some to the low tens of thousands. This changes the interpretation of the headline: a +50k print can be consistent with a stable labour market.

2. What It Implies & How It Affects Macro Strategy

The biggest monthly market mover

Payrolls set the tone for rates, FX and equities because the labour market is central to the Fed's dual mandate and to consumer income. A typical reaction function: strong payrolls + hot wages = higher yields, stronger dollar, pressure on duration-sensitive equities; weak payrolls = lower front-end yields, steeper curve, weaker dollar. In late-cycle phases, 'bad news' becomes bad for equities as recession fears dominate rate-cut hopes.

How strategists read it

- Use the **3-month average** including revisions rather than the headline; compare with the breakeven pace.
- **Breadth:** the diffusion index and the share of job gains from a few sectors (e.g. health care and government 'acyclical' hiring) show whether demand is broad.
- **Wages:** average hourly earnings y/y around 3-3.5% is roughly consistent with 2% inflation given ~1-1.5% productivity growth; hotter wages imply sticky services inflation. Cross-check with the Employment Cost Index (quarterly) and Atlanta Fed Wage Growth Tracker.
- **Hours:** falling average weekly hours often precede layoffs (firms cut hours first). Aggregate weekly payrolls (jobs x hours x wages) proxy labour income.
- **Confirm with other labour data:** weekly initial and continuing jobless claims (DOL, Thursdays), JOLTS openings/quits/hires, ADP private payrolls (Wednesday before NFP), Challenger layoffs and the household survey.

3. How to Get the Data Free

Exhibit 18: Where to get payroll and labour data for free

Source	What you get	Notes
bls.gov/ces and bls.gov/news.release/empsit	Full Employment Situation release, tables B-1 to B-9, revisions, benchmark notices	Release calendar at bls.gov/schedule
BLS API	Series CES0000000001 (total non-farm level), CES0500000003 (AHE)	Free key
FRED / ALFRED	PAYEMS, USPRIV, CES0500000003, AWHAETP, TEMPHELPS; ALFRED for first-release vintages	Use 'Change, thousands' transform
QCEW (bls.gov/cew)	Quarterly near-census counts by county/industry	Earliest check on benchmark revisions
DOL (dol.gov/ui/data.pdf); FRED ICSA, CCSA	Weekly initial and continuing claims	Highest-frequency labour data
ADP (adpemploymentreport.com)	ADP National Employment Report	Free; independent sample

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 9 | LABOUR MARKET

Unemployment Rate

Bottom line

- The Fed's main gauge of maximum employment; compare with u* (~4.0-4.3%).
- Turns late, then rises non-linearly: the **Sahm rule** captures this, but the cause of the rise matters.
- Monthly changes of 0.1pp are within sampling error.

At a glance

Indicator type	Hard data (labour)
Lead / lag	Lagging (turns late)
Key question	Labour-market slack; the Fed's maximum-employment mandate; recession rules
Publisher	BLS, from the Current Population Survey (CPS), released in the same Employment Situation report as payrolls
Frequency	Monthly, first Friday, 8:30 ET
Headline	U-3: unemployed as a % of the civilian labour force (SA), reported to one decimal
FRED codes	UNRATE (U-3), U6RATE (U-6), CIVPART (participation), LNS12300060 (prime-age employment-population ratio), SAHMREALTIME (Sahm rule), NROU (CBO natural rate)
Taipei time	21:30 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

The CPS (or 'household survey') is run by the Census Bureau for BLS and interviews roughly 60,000 eligible households each month (sample sizes have been under budget pressure). Respondents are classified, for the week including the 12th, as **employed**, **unemployed** (no job, available, and actively searched in the past 4 weeks or on temporary layoff) or **not in the labour force**.

Exhibit 19: Labour underutilisation measures

Measure	Definition
U-3 (headline)	Unemployed / labour force
U-1, U-2	Long-term unemployed (15+ weeks); job losers
U-4	U-3 + discouraged workers
U-5	U-4 + other marginally attached workers
U-6	U-5 + part-time for economic reasons (underemployment); broadest slack measure
Participation rate	Labour force / civilian non-institutional population 16+ (falls with ageing)
Employment-population ratio (prime age 25-54)	Avoids ageing distortions; many economists' preferred slack gauge

Source: BLS.

Quirks

- **Sampling error:** the 90% confidence interval for the monthly change in U-3 is about +/-0.2pp, so 0.1pp moves are statistical noise.
- **Population controls:** each January BLS updates population estimates (immigration revisions can shift levels by millions), creating breaks in household-survey levels but little in the rate.
- **Household vs establishment:** household employment counts people (incl. self-employed) and can diverge from payrolls for months.
- **Labour-supply effects:** slower immigration can keep the rate low even with weak hiring - the labour market can be 'low hire, low fire' with a stable rate.

2. What It Implies & How It Affects Macro Strategy

Mandate and the Phillips curve

The unemployment rate is the Fed's primary gauge of 'maximum employment'. The FOMC's SEP gives a longer-run estimate of the natural rate (u*, roughly 4.0-4.3% in recent projections). Unemployment well below u* implies wage-driven inflation risk; above u* implies slack and justifies easier policy.

A powerful recession signal - because it moves late and then fast

- The unemployment rate tends to be stable or falling through expansions and then rises non-linearly: firms cut hiring, then lay off, then income falls, then spending falls - a self-reinforcing loop.
- **Sahm Rule** (Claudia Sahm): a recession signal is triggered when the 3-month moving average of U-3 rises 0.5pp or more above its minimum over the prior 12 months. It had a near-perfect real-time record until it triggered in mid-2024 without a recession (largely due to rising labour supply), showing that the cause of the rise matters: new entrants and immigrants vs permanent job losers.
- Watch the composition: **permanent job losers**, continuing claims and long-term unemployment are the 'bad' type of increase; re-entrants are less worrying.

Market implications

- Rising unemployment: lower front-end yields and steeper curve (bull steepening), weaker dollar against funding currencies, wider credit spreads, underperformance of consumer cyclicals and small caps; equities fall if earnings risk outweighs rate cuts.
- Falling or very low unemployment with sticky inflation: higher-for-longer policy, flatter curve, value and financials may outperform.
- The unemployment rate and payrolls together give the policy read: e.g. weak payrolls but a stable rate (low labour supply growth) is less dovish than it looks.

3. How to Get the Data Free

Exhibit 20: Where to get unemployment data for free

Source	What you get	Notes
bls.gov/cps	Tables A-1 to A-16 (incl. U-1 to U-6 in Table A-15), demographic breakdowns, flows	CPS microdata available via Census/IPUMS (ipums.org) for free research
FRED	UNRATE, U6RATE, CIVPART, LNS12300060, SAHMREALTIME, NROU	SAHMREALTIME is computed from real-time vintages
FOMC SEP (federalreserve.gov)	Fed's projections for unemployment and longer-run u*	Quarterly with FOMC meetings
State data: BLS LAUS	State and metro unemployment rates	Regional divergence

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 10 | LABOUR MARKET

Average Wages: Average Hourly Earnings, Employment Cost Index & Wage Trackers

Bottom line

- **ECI** is the Fed's preferred wage gauge because it holds the job mix fixed; AHE is timelier but distorted by composition.
- Wage growth of roughly 3-3.5% is consistent with 2% inflation if productivity grows ~1-1.5%.
- The Atlanta Fed tracker separates job-switchers from job-stayers - a read on labour-market bargaining power.

At a glance

Indicator type	Hard data (labour costs)
Lead / lag	Lagging
Key question	Is wage growth consistent with 2% inflation?
Publishers	BLS (Average Hourly Earnings in the Employment Situation; Employment Cost Index); Federal Reserve Bank of Atlanta (Wage Growth Tracker)
Frequency	AHE monthly with payrolls (first Friday, 8:30 ET); ECI quarterly, end of the month after each quarter (late Jan/Apr/Jul/Oct, 8:30 ET); Atlanta Fed tracker monthly, after the CPS microdata
Headline	AHE m/m and y/y; ECI q/q (SA) and y/y for total compensation and wages & salaries
FRED codes	CES0500000003 (AHE, all private), AHETPI (production & non-supervisory), ECIWAG (ECI wages & salaries, private), ECIALLCIV (ECI total compensation), PRS85006112 (unit labour costs)
Taipei time	AHE and ECI 21:30 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

Exhibit 21: Wage measures compared

Measure	How it is built	Strengths	Weaknesses
Average Hourly Earnings (AHE)	Total payroll / total hours from the CES establishment survey	Monthly, timely, large sample	**Composition effects**: in 2020 AHE spiked because low-wage workers lost jobs; excludes benefits
Employment Cost Index (ECI)	Fixed-weight (Laspeyres) index from the National Compensation Survey; the same jobs tracked over time across occupations and industries	Controls for job mix; includes benefits; the **Fed's preferred** wage gauge	Quarterly and released late; private wages ex-incentive-paid occupations is the cleanest series
Atlanta Fed Wage Growth Tracker	Median 12-month wage change for the same individuals observed a year apart in the Current Population Survey (3-month moving average)	Person-level; splits **job-switchers vs job-stayers**, by age, wage quartile	Noisy at monthly frequency; based on household survey sample
Unit labour costs	Compensation per hour / output per hour (BLS productivity release)	Links wages to productivity - what matters for inflation	Quarterly, heavily revised
Private alternatives	ADP Pay Insights; Indeed Wage Tracker (posted wages)	Timely, large samples	Different coverage and methods

Source: BLS, Federal Reserve Bank of Atlanta.

2. What It Implies & How It Affects Macro Strategy

Wages and the inflation mandate

Labour is the largest cost in services, so wage growth anchors **core services inflation**. A useful rule: sustainable nominal wage growth = target inflation (2%) + trend productivity growth (~1-1.5%), i.e. roughly **3-3.5%**. Wage growth well above that, with weak productivity, implies sticky services inflation and a hawkish Fed; wage growth near or below it gives the Fed room to ease.

How strategists use it

- **AHE on payroll day:** a 0.1pp surprise in AHE m/m moves front-end yields alongside the payroll headline; check average weekly hours, because composition and hours shifts distort the series.
- **ECI as the anchor:** the Fed cites ECI to judge whether labour costs are consistent with 2%; a hot ECI print in late January or April can reset rate expectations for a quarter.
- **Switcher premium:** when job-switchers' wage growth far exceeds job-stayers', the labour market is tight; a narrowing gap (as in 2023-25) signals reduced worker bargaining power and cooling wage pressure.
- **Real wages:** wage growth minus CPI inflation drives real income and consumption; positive real wage growth supports consumer spending even with slower hiring.
- **Margins:** wages growing faster than prices squeeze corporate margins, especially in labour-intensive sectors (restaurants, retail, healthcare).

3. How to Get the Data Free

Exhibit 22: Where to get wage data for free

Source	What you get	Notes
bls.gov/ces; bls.gov/eci	AHE by industry; ECI releases and full history by occupation, industry, union status	BLS API for programmatic access
atlantafed.org (Wage Growth Tracker)	Overall median and breakdowns (switchers/stayers, age, wage level, industry); downloadable Excel	Updated monthly
FRED	CES0500000003, AHETPI, ECIWAG, ECIALLCIV, PRS85006112	Use 'Percent change from year ago'
ADP; Indeed Hiring Lab	Pay Insights; posted-wage tracker	Free

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 11 | LABOUR MARKET

JOLTS: Job Openings and Labor Turnover Survey

Bottom line

- Openings, hires, **quits** and layoffs show labour-market flows behind the payroll headline.
- The quits rate leads wage growth; the openings-per-unemployed ratio measures tightness.
- Lags 5-6 weeks and is heavily revised: trade the trend, not the headline openings number.

At a glance

Indicator type	Hard data (labour flows)
Lead / lag	Leading (quits, openings)
Key question	How tight is the labour market, and is churn slowing?
Publisher	BLS
Frequency	Monthly, ~5-6 weeks after the reference month (usually the Tuesday after payrolls or the following week), 10:00 ET
Headline	Job openings level (millions) and rate; hires, quits, layoffs & discharges, total separations
FRED codes	JTSJOL (openings), JTSHIR (hires), JTSQUR (quits rate), JTSLDL (layoffs & discharges), UNEMPLOY (to compute openings per unemployed)
Taipei time	23:00 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

JOLTS surveys roughly 21,000 establishments about job openings and labour turnover. A **job opening** requires that a specific position exists, work could start within 30 days, and the employer is actively recruiting from outside the establishment. Hires and separations are counted over the whole month; openings are measured on the last business day. Estimates are aligned with CES payroll employment, and the survey has a relatively low response rate, so monthly openings figures carry wide error bands and are frequently revised.

Exhibit 23: JOLTS components and what they signal

Component	Definition	Macro signal
Job openings	Unfilled positions actively recruited	Labour demand; peaked above 12 million in 2022
Hires	Additions to payroll during the month	Actual hiring activity; a 'low-hire' market makes it hard for the unemployed to find jobs
Quits	Voluntary separations	Worker confidence; the **quits rate leads wage growth** by several months
Layoffs & discharges	Involuntary separations	Stress indicator; stayed low through 2023-25 ('low hire, low fire')
Openings / unemployed (V/U)	Openings divided by number of unemployed	Tightness: ~2.0 in 2022 vs about 1.0 or below by 2025, near pre-pandemic balance

Source: BLS.

2. What It Implies & How It Affects Macro Strategy

Reading the flows behind the headline

Payrolls show the **net** change in jobs; JOLTS shows the gross flows underneath. Roughly 5 million hires and 5 million separations each month produce a net change in the tens of thousands. A labour market can therefore weaken through falling hires and quits long before layoffs rise. The **Beveridge curve** (openings rate vs unemployment rate) was central to the 2022-24 'soft landing' debate: openings fell sharply with little rise in unemployment, which suggested the economy could cool without mass layoffs.

How strategists use it

- **Wage outlook:** a falling quits rate signals less job-hopping and slower wage growth ahead - a dovish signal for services inflation.
- **Recession risk:** rising layoffs plus falling hires is the dangerous combination; watch it with weekly jobless claims and continuing claims.
- **Fed communication:** the Fed has cited the V/U ratio as a measure of labour-market tightness; a ratio near 1 is consistent with a balanced market.
- **Market reaction:** openings surprises can move rates, but because of revisions and the lag, treat JOLTS as a trend confirmation; Indeed's daily job-postings index offers a timelier alternative.

3. How to Get the Data Free

Exhibit 24: Where to get JOLTS data for free

Source	What you get	Notes
bls.gov/jlt	Release, tables by industry and region, rates and levels, revisions	BLS API series JTS...
FRED	JTSJOL, JTSHIR, JTSQUR, JTSLDL; combine with UNEMPLOY for V/U	Graph builder can compute ratios
Indeed Hiring Lab (hiringlab.org)	Daily job-postings index by occupation and country	Free CSV on GitHub

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 12 | LABOUR MARKET

ADP National Employment Report

Bottom line

- An independent, payroll-records-based estimate of private employment released two days before NFP.
- Month-to-month correlation with BLS private payrolls is loose: treat it as directional.
- ADP Pay Insights (job-stayers vs job-changers) is a useful complement to AHE and ECI.

At a glance

Indicator type	Private-sector data (payroll records)
Lead / lag	Coincident
Key question	A directional preview of private payrolls and pay growth
Publisher	ADP Research, in collaboration with the Stanford Digital Economy Lab
Frequency	Monthly, usually the Wednesday before the BLS jobs report, 8:15 ET; ADP has also begun publishing a weekly preliminary estimate of private employment
Headline	Monthly change in US private employment (thousands); ADP Pay Insights median pay change for job-stayers and job-changers
FRED codes	ADP series are carried on FRED (search 'ADP'); primary source adpemploymentreport.com
Taipei time	21:15 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

ADP processes payroll for a large share of US employers. The report uses anonymised payroll records covering millions of private-sector workers to estimate monthly employment change, weighted to match the structure of US employment by industry and firm size. ADP overhauled the methodology in 2022 with Stanford economists: it now counts **employees who were paid** during the pay period including the 12th of the month (aligned with BLS timing) and no longer models the estimate on BLS data. The report breaks down jobs by industry, firm size and region.

Pay Insights

Because ADP tracks the same individuals over time, it reports median annual pay growth for **job-stayers** and **job-changers** - similar in spirit to the Atlanta Fed tracker but from administrative payroll data rather than a survey.

2. What It Implies & How It Affects Macro Strategy

A directional preview, not a forecast

Markets read ADP as a preview of the private-payrolls component of the BLS report, but the month-to-month correlation is loose: differences in sample, definitions and seasonal adjustment mean the two can diverge by 100k or more in a given month. ADP is most useful for **trend and turning points**, and it gains importance when official data are delayed or disrupted (as during the 2025 government shutdown).

How strategists use it

- **Positioning ahead of NFP:** a large ADP surprise shifts expectations for Friday and can move front-end yields on Wednesday, but the effect usually fades once BLS data arrive.
- **Small-business read:** ADP's firm-size breakdown shows whether small firms (most sensitive to credit conditions) are hiring or cutting.
- **Wage pressure:** a narrowing gap between job-changer and job-stayer pay growth confirms cooling labour demand.
- **Cross-check:** combine with weekly jobless claims, the Conference Board labour differential, Challenger layoff announcements and ISM employment indices for a fuller pre-NFP picture.

3. How to Get the Data Free

Exhibit 25: Where to get ADP data for free

Source	What you get	Notes
adpemploymentreport.com	Monthly report, historical data download, methodology, Pay Insights	Free
FRED	ADP private employment series	Search 'ADP' on FRED
Economic calendars	Consensus and prior for ADP and NFP	Investing.com, ForexFactory

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 13 | INFLATION

CPI / Core CPI

Bottom line

- **Core CPI m/m to two decimals** is usually the most market-moving inflation number.
- Decompose into core goods, shelter (lags market rents by ~9-12 months) and services ex-shelter.
- CPI components let economists nowcast core PCE, the Fed's target, two weeks early.

At a glance

Indicator type	Hard data (prices)
Lead / lag	Coincident / lagging
Key question	The inflation trend; the main driver of rate expectations
Publisher	US Bureau of Labor Statistics (BLS), Department of Labor
Frequency	Monthly, usually ~10th-15th of the following month, 8:30 ET
Headline series	CPI-U all items (headline); CPI-U less food and energy (core); m/m SA and y/y NSA
FRED codes	CPIAUCSL (headline SA index), CPILFESL (core SA), CUSR0000SAH1 (shelter), CPIAUCNS (NSA)
Taipei time	21:30 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

The Consumer Price Index measures the average change over time in prices paid by urban consumers for a fixed basket of goods and services. BLS data collectors (in person, by phone and increasingly online/via corporate data) record roughly 80,000+ prices each month from tens of thousands of retail and service establishments in ~75 urban areas, plus rents from a separate housing survey of ~50,000 rental units.

Construction

- **Basket weights** come from the Consumer Expenditure Survey. Since 2023 weights are updated **annually** (using a single year of data, lagged two years) rather than every two years.
- **Two-stage aggregation:** within each of ~240 item strata in each area, prices are combined with a **geometric mean** (which allows for some substitution within categories); strata are then combined into the overall index with a modified **Laspeyres** formula (fixed weights).
- **Quality adjustment:** when products change (new phone models, apparel), BLS uses hedonic regression and other methods to separate pure price change from quality change.
- **Shelter** (~35% of headline, ~44% of core) is measured via rents. Homeowners are imputed an **Owners' Equivalent Rent (OER)**: what they would pay to rent their own home. House prices and mortgage rates are not in CPI. Because leases reset slowly, CPI shelter lags market rents (Zillow, Apartment List, the BLS/Cleveland Fed New Tenant Rent index) by about 9-12 months.

Variants

Exhibit 26: CPI variants and alternative trend measures

Measure	Definition	Use
CPI-U (headline)	All urban consumers (~90%+ of population)	Market headline
Core CPI	CPI-U excluding food and energy	Underlying trend; drives rates
CPI-W	Urban wage earners and clerical workers	Social Security COLA indexation
C-CPI-U (chained)	Superlative formula capturing substitution; revised	Tax-bracket indexation; runs ~0.25pp below CPI-U
'Supercore'	Core services excluding shelter (unofficial, but published components let you build it)	Labour-intensive services inflation; Fed focus 2022-24
Core goods	Commodities less food and energy	Tariff, supply-chain and dollar effects

Measure	Definition	Use
Median / Trimmed-mean CPI	Cleveland Fed: removes extreme component moves each month	Robust trend
Sticky-price CPI	Atlanta Fed: items whose prices change infrequently	Embedded inflation expectations

Source: BLS, Federal Reserve Banks of Cleveland and Atlanta.

Data-quality caveats

During the 2025 federal government shutdown BLS was unable to collect October 2025 prices, leaving a gap in the CPI series, and staffing constraints in 2025 increased the share of prices imputed rather than collected. When using 2025-26 data, check BLS notices for imputation rates and any missing months, and prefer multi-month averages.

2. What It Implies & How It Affects Macro Strategy

The most market-moving inflation print

Although the Fed targets PCE, CPI is released about two weeks earlier and its components allow economists to nowcast PCE with good accuracy. Hence **core CPI m/m (to two decimals)** is typically the single most important inflation number for the rates market. A 0.1pp surprise in core m/m can move 2-year Treasury yields by 10-20bp in a sensitive regime. As a rule of thumb, core CPI averaging roughly 0.17% m/m annualises to about 2%.

How strategists read it

- **Decompose:** core goods (tariffs, used cars, supply chains), shelter (lagging, predictable from market rents), supercore services (wages and demand). Investors care most about whether the trend in services ex-shelter is consistent with 2%.
- **Translate to PCE:** after CPI and PPI, sell-side economists publish core PCE estimates; the market reacts to the implied PCE as much as to CPI itself.
- **Breadth:** the share of components rising above a 3% or 4% annualised pace distinguishes a broad inflation problem from a few idiosyncratic items (eggs, airfares, auto insurance).
- **Real yields and TIPS:** CPI-U NSA is the index that accrues to TIPS principal, so CPI prints feed directly into breakevens and TIPS carry. CPI swap 'fixings' trade on future prints.
- **Policy reaction function:** hot core prints raise terminal-rate pricing, flatten the curve, strengthen the dollar and hurt long-duration equities (Nasdaq, unprofitable growth, small caps); soft prints do the reverse.
- **Real income:** CPI deflates wages (real earnings), retail sales and household income - weak real income growth is a consumption headwind.

> **Practical checklist on CPI day**
>
> 1) Core m/m vs consensus (to two decimals), 2) supercore and shelter m/m, 3) core goods (tariff pass-through), 4) 3m and 6m annualised core, 5) implied core PCE estimate, 6) revisions are rare in CPI except seasonal-factor updates each February.

3. How to Get the Data Free

Exhibit 27: Where to get CPI data for free

Source	What you get	Notes
bls.gov/cpi	Release PDF, detailed tables (relative importance weights, 200+ components), historical data, methodology handbook	'Data Tools' > 'Top Picks' or 'One-Screen' to build CSVs
BLS Public Data API (api.bls.gov)	Programmatic access; series e.g. CUSR0000SA0 (headline SA), CUSR0000SA0L1E (core SA)	Free key raises limits (v2: 500 queries/day, 20 years per query)
FRED	CPIAUCSL, CPILFESL, CUSR0000SASLE (services less energy), MEDCPIM158SFRBCLE (median CPI)	Best for charts, transformations, Excel add-in
Cleveland Fed	Median/trimmed-mean CPI and the daily **Inflation Nowcasting** model for CPI and PCE	clevelandfed.org
Atlanta Fed	Sticky/flexible CPI; Business Inflation Expectations	atlantafed.org

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 14 | INFLATION

PPI / Core PPI: Producer Price Index

Bottom line

- Measures prices received by **domestic producers**; headline is the Final Demand index (goods, services, construction).
- Its biggest market role is feeding **core PCE**: portfolio management, airfares and health-care prices come from PPI.
- Trade-services margins and intermediate-demand indices show pipeline pressure and tariff pass-through.

At a glance

Indicator type	Hard data (prices)
Lead / lag	Coincident / leading for goods prices
Key question	Cost pressure in the production pipeline, margins, and the PPI inputs to core PCE
Publisher	US Bureau of Labor Statistics (BLS)
Frequency	Monthly, usually within a day or two of CPI (around the 10th-15th of the following month), 8:30 ET; data are revised four months after first publication
Headline	PPI for Final Demand, m/m (SA) and y/y (NSA); core = final demand less foods and energy, and less foods, energy and trade services
FRED codes	PPIFIS (final demand), PPIACO (all commodities, legacy index); search FRED for 'final demand less foods, energy' for the core series
Taipei time	21:30 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

The Producer Price Index measures the average change over time in the **selling prices received by domestic producers** for their output - the first commercial transaction - rather than prices paid by consumers. BLS collects tens of thousands of price quotes each month from a probability sample of establishments across goods, services and construction. Prices exclude sales and excise taxes, and the index covers roughly three-quarters of US services output as well as goods.

Final Demand-Intermediate Demand (FD-ID) structure

Since 2014 the headline is the **Final Demand** index, which covers goods, services and construction sold for personal consumption, capital investment, government and export. A parallel set of **Intermediate Demand** indices tracks prices of inputs to production, organised by stage of processing, which lets analysts follow price pressure through the supply chain.

Exhibit 28: PPI Final Demand components

Component	Approx. weight	Notes
Final demand goods	~1/3	Foods, energy, and core goods (consumer goods and capital equipment); energy makes the headline volatile
Final demand services	~2/3	Includes **trade services** (wholesale and retail margins), transportation and warehousing, and other services such as portfolio management, airline passenger services and health care
Trade services (within services)	~1/5 of final demand	Measured as **margins** (selling price minus acquisition cost). Margins swing when retailers absorb or pass on cost changes, so BLS also publishes a core excluding trade
Final demand construction	Small	Non-residential buildings sold to final users
Intermediate demand (stages 1-4)	Separate system	Prices of processed goods, unprocessed goods and services used as inputs; earlier stages lead later stages

Source: BLS; shares approximate.

Core measures

- **Final demand less foods and energy** - the traditional core.

- **Final demand less foods, energy and trade services** - BLS's narrowest core; strips out volatile retail and wholesale margins and is often the cleaner underlying trend.
- Revisions four months after first publication can be meaningful, and monthly PPI is more volatile than CPI.

2. What It Implies & How It Affects Macro Strategy

From producer prices to consumer prices

PPI is often described as a leading indicator of CPI, but the link is looser than it sounds: producer costs pass through to consumers with variable lags, margins absorb part of the change, and imports (not in PPI) and shelter (dominant in CPI) break the chain. PPI is most useful for **goods inflation** - core goods PPI and intermediate demand indices lead core goods CPI by a few months - and for spotting **tariff pass-through** and margin squeezes.

The PCE connection - why markets care

Several PCE components are taken from PPI rather than CPI, including **portfolio management and investment advice** (which move with stock prices), **airline passenger services**, and **physician, hospital and other medical services**. After CPI and PPI are both out, economists translate them into a precise estimate of core PCE, the Fed's target measure. As a result a 'hot' PPI headline can be read as dovish if its PCE-relevant components are soft, and vice versa - the market reaction often depends on the implied core PCE, not the PPI headline.

How strategists use it

- **Implied core PCE:** combine CPI and PPI components to update the core PCE estimate; this is the main rates-market use of PPI.
- **Margins and earnings:** PPI rising faster than CPI signals margin pressure for retailers and consumer-goods companies; trade-services margins show whether retailers are absorbing tariffs or passing them on.
- **Pipeline pressure:** intermediate-demand indices and ISM Prices Paid together flag whether upstream cost pressure is building or fading.
- **Tariffs and the dollar:** import-sensitive goods categories, together with the BLS Import/Export Price Indexes, show who bears tariff costs - foreign exporters, US producers or consumers.
- **Cross-check with surveys:** S&P Global Input Prices and Output Prices indices give an earlier read on the same pressures.

3. How to Get the Data Free

Exhibit 29: Where to get PPI data for free

Source	What you get	Notes
bls.gov/ppi	Release, detailed tables for final and intermediate demand, thousands of industry and commodity indices, methodology	Data Tools > Top Picks for core series
BLS Public Data API	Programmatic access to PPI series (series IDs start with WP for commodity-based indices)	Free key
FRED	PPIFIS, PPIACO, and hundreds of detailed PPI series	Search by component name
bls.gov/mxp	Import and Export Price Indexes - the companion series for traded goods	Monthly

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 15 | INFLATION

PCE / Core PCE Price Index

Bottom line

- The FOMC's 2% target is defined in **PCE**; core PCE is the trend gauge in Fed projections.
- Runs ~0.3-0.5pp below CPI due to formula, weight (lower shelter, higher healthcare) and scope effects.
- Rarely surprises because it is nowcast from CPI/PPI; real spending and revisions often matter more.

At a glance

Indicator type	Hard data (prices)
Lead / lag	Coincident / lagging
Key question	Distance from the Fed's 2% target (the official target measure)
Publisher	US Bureau of Economic Analysis (BEA), Department of Commerce
Frequency	Monthly, in the 'Personal Income and Outlays' release, typically late in the following month, 8:30 ET
Policy role	The FOMC's 2% inflation objective is defined in terms of **headline PCE** (since 2012); **core PCE** is the main gauge of the underlying trend
FRED codes	PCEPI (headline), PCEPILFE (core), PCETRIM12M159SFRBDAL (Dallas Fed trimmed mean)
Taipei time	21:30 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

The PCE price index is the deflator for Personal Consumption Expenditures in the national accounts. BEA largely reuses CPI component prices but supplements them with PPI data (e.g. airfares, medical services, hospital prices) and other sources (e.g. financial-services prices imputed from asset values). The index is a **chain-type Fisher** index with weights updated every period.

Why PCE differs from CPI (typically runs ~0.3-0.5pp lower)

Exhibit 30: Why PCE inflation differs from CPI

Effect	CPI	PCE
Formula effect	Fixed-weight Laspeyres (upper level)	Chain-weighted Fisher: captures substitution between categories as relative prices change
Weight effect	Household survey (out-of-pocket spending). Shelter ~35%	Business data from national accounts. Shelter ~15%; health care much larger
Scope effect	Out-of-pocket spending by urban consumers only	All spending by and on behalf of households, incl. employer- and government-paid health care (Medicare/Medicaid) and non-profits
Other	-	Different seasonal adjustment, price sources for some items (PPI), and revisions

Source: BEA, BLS.

Revisions and variants

- PCE is **revised** with subsequent monthly releases and in annual GDP updates; CPI is essentially not revised.
- **Core PCE** excludes food and energy. **Market-based PCE** excludes imputed prices (e.g. financial service charges, which move with stock prices - a reason PCE can surprise when equities rally).
- **Dallas Fed trimmed-mean PCE** drops the most extreme price changes each month - a favourite Fed-staff trend measure.
- The same release contains **personal income, spending (nominal and real), the saving rate and disposable income**, which are major consumption indicators in their own right.

2. What It Implies & How It Affects Macro Strategy

The Fed's yardstick

Because PCE is the official target, the FOMC's Summary of Economic Projections (SEP) forecasts headline and core PCE, and Fed communications cite it. However, market reaction on PCE day is often muted because the number is **well anticipated** from CPI and PPI components; the surprise versus the updated nowcast is what matters. The bigger market-moving content can be real spending and income, or large revisions.

How strategists use it

- **Distance from target:** the gap between core PCE (3m/6m annualised and y/y) and 2% defines the Fed's room to cut or need to hike. Compare to SEP projections to see whether the Fed is likely to be surprised.
- **Real policy rate:** fed funds minus core PCE (or minus expected inflation) approximates the real policy stance. Comparing it with estimates of the neutral rate (r*, e.g. NY Fed Holston-Laubach-Williams) shows whether policy is restrictive.
- **Healthcare and financial services:** because these have larger weights in PCE than CPI, Medicare reimbursement changes and stock market moves (portfolio-management fees) can make PCE diverge from CPI.
- **Consumption momentum:** real PCE spending is ~68% of GDP; the monthly path drives GDP nowcasts (GDPNow updates on this release).
- **Saving rate:** a low saving rate suggests limited buffers (consumption vulnerable to shocks); a high rate suggests pent-up spending power.

3. How to Get the Data Free

Exhibit 31: Where to get PCE data for free

Source	What you get	Notes
bea.gov > Personal Income and Outlays	Release, full tables (NIPA Table 2.8.7 for price changes, 2.3.x for spending)	Interactive Data Application lets you download any NIPA table
BEA API (apps.bea.gov/API)	Programmatic NIPA access	Free registration key
FRED	PCEPI, PCEPILFE, PCE (nominal), PCEC96 (real), PSAVERT (saving rate)	Use 'Edit Graph' for percent change transforms
Dallas Fed	Trimmed-mean PCE (1, 6, 12-month)	dallasfed.org
Cleveland Fed nowcast	Daily nowcasts of current-month PCE and core PCE	Useful pre-release

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 16 | MONETARY POLICY & RATES

Central Bank Interest Rate: the Fed Funds Rate & the FOMC

Bottom line

- The FOMC sets a **target range** for the fed funds rate at eight scheduled meetings a year.
- The statement, press conference, quarterly **dot plot** and minutes reveal the reaction function.
- What matters for markets is the decision and guidance relative to what futures already price.

At a glance

Indicator type	Policy decision
Lead / lag	Sets the price of money
Key question	Where policy is, where the FOMC thinks it is going, and why
Publisher	Federal Open Market Committee (FOMC), Federal Reserve System; effective rates published by the Federal Reserve Bank of New York
Frequency	Eight scheduled meetings a year (plus unscheduled meetings if needed). Statement 2:00 p.m. ET, Chair's press conference 2:30 p.m. ET; Summary of Economic Projections (dot plot) in Mar/Jun/Sep/Dec; minutes three weeks after each meeting
Headline	Target range for the federal funds rate (25bp wide), plus policy statement and guidance
FRED codes	DFEDTARU / DFEDTARL (target range), DFF / FEDFUNDS (effective fed funds rate), IORB (interest on reserve balances), SOFR, WALCL (Fed balance sheet)
Taipei time	Statement 03:00, press conference 03:30 (next day) (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

The federal funds rate is the rate at which banks lend reserves to each other overnight, unsecured. The FOMC - seven Board governors plus five of the twelve regional Reserve Bank presidents (New York always votes; the other four rotate annually) - sets a **target range**. Because the banking system holds abundant reserves, the Fed steers the market rate with **administered rates** rather than daily reserve scarcity:

Exhibit 32: How the Fed implements its target range

Tool / rate	Role
Interest on reserve balances (IORB)	Paid to banks on reserves at the Fed; banks will not lend below it, so it anchors the effective rate inside the range
Overnight reverse repo (ON RRP) rate	Set at the bottom of the range; money-market funds lend to the Fed at this rate, creating a floor for short-term rates
Standing repo facility (SRF)	Lets eligible institutions borrow against Treasuries near the top of the range; a ceiling for repo rates in stress
Effective fed funds rate (EFFR)	Volume-weighted median of overnight fed funds trades, published daily by the New York Fed (usually a few bp above the bottom of the range)
SOFR	Secured Overnight Financing Rate - the broad Treasury repo rate; the main benchmark for derivatives and floating-rate loans since LIBOR's demise
Balance sheet	Asset purchases (QE) and runoff (QT) affect long-term yields and reserve levels; the Fed ended QT at the end of 2025

Source: Federal Reserve Board, Federal Reserve Bank of New York.

The communication package

- **Statement:** decision, vote (dissents matter), and assessment of growth, labour and inflation; small wording changes are parsed closely.
- **Press conference:** the Chair's answers often move markets more than the statement itself.

- **Summary of Economic Projections (SEP):** each participant's projections for GDP, unemployment, PCE and core PCE inflation, plus the **dot plot** of appropriate year-end policy rates and the longer-run neutral rate.
- **Minutes** (3 weeks later), Fed speeches, the semi-annual Monetary Policy Report and testimony, and the Beige Book (anecdotal regional conditions, eight times a year) fill in the reaction function.
- **Blackout period:** officials stop public comments from the second Saturday before each meeting.

2. What It Implies & How It Affects Macro Strategy

The price of money for everything

The policy rate anchors short-term rates, which feed into Treasury yields, mortgage and credit card rates, corporate borrowing, equity valuations and the dollar. Policy works with long and variable lags (often 12-24 months). The key concepts are the **neutral rate** (r* in real terms, the rate that neither stimulates nor restrains; FOMC longer-run dots imply roughly 3% nominal) and the **real policy rate** (fed funds minus expected or core inflation).

How strategists use it

- **Surprise vs pricing:** the FOMC rarely surprises on the decision itself; the market impact comes from the dots, statement language and press conference relative to what futures price (see the CME FedWatch chapter).
- **Reaction function:** map each data release onto the dual mandate - inflation (core PCE vs 2%) and maximum employment (unemployment vs u*). Simple rules (Taylor rule variants published by the Atlanta and Cleveland Feds) give a benchmark for where policy 'should' be.
- **FX:** rate differentials versus the ECB, BoJ and others drive the dollar; a dovish Fed relative to peers weakens the dollar and supports Asian currencies including the TWD.
- **Cutting cycles:** history distinguishes 'insurance' cuts during expansions (equities usually rally) from recession cuts (equities fall first); the context matters more than the cut.
- **Independence and leadership:** perceived political pressure on the Fed tends to raise inflation expectations and the term premium; leadership changes (the Chair's term is four years) bring reaction-function uncertainty.

3. How to Get the Data Free

Exhibit 33: Where to get Fed policy data for free

Source	What you get	Notes
federalreserve.gov	FOMC calendar, statements, press-conference transcripts and video, SEP, minutes, Beige Book, H.15 interest rates, H.4.1 balance sheet	Implementation notes give the exact administered rates
newyorkfed.org (Markets)	Daily EFFR, OBFR, SOFR, repo operations; Survey of Market Expectations; r* estimates (Holston-Laubach-Williams)	CSV and API
FRED	DFEDTARU, DFEDTARL, DFF, FEDFUNDS, IORB, SOFR, WALCL	Daily and monthly
Atlanta Fed / Cleveland Fed	Taylor rule utility; market-based probability tools	Free

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 17 | MONETARY POLICY & RATES

US Treasury Yields (2Y / 5Y / 10Y / 20Y / 30Y)

Bottom line

- The **2-year** tracks expected Fed policy; the **10- and 30-year** add inflation compensation and term premium.
- Curve inversions (2s10s, 3m10y) have preceded every US recession since the 1960s, with long and variable lags.
- Treasury yields are the discount rate for every other asset and set mortgage and corporate borrowing costs.

At a glance

Indicator type	Market price
Lead / lag	Real time (yield curve is leading)
Key question	Expected policy path, inflation compensation, term premium and recession signals
Publisher	US Department of the Treasury (Daily Treasury Par Yield Curve Rates); Federal Reserve H.15
Frequency	Markets trade nearly 24 hours; Treasury publishes official par yields each business day based on closing indications; auctions follow a published schedule
Headline	Yields to maturity on benchmark notes and bonds; key curve spreads (2s10s, 3m10y, 5s30s)
FRED codes	DGS2, DGS5, DGS10, DGS20, DGS30, DGS3MO; T10Y2Y, T10Y3M (spreads); DFII10 (10y TIPS real yield); T10YIE (10y breakeven inflation); THREEFYTP10 (Kim-Wright 10y term premium)
Taipei time	Trades nearly 24h; Treasury's par curve is published after the New York close

1. Methodology

Treasury's **par yield curve** is derived from indicative bid-side prices of the most recently auctioned ('on-the-run') securities at about 3:30 p.m. ET, fitted with a monotone convex spline to give constant-maturity yields from 1 month to 30 years (the 20-year bond returned in 2020). Yields move inversely to prices. Benchmark futures trade at CME (2Y, 5Y, 10Y, Ultra 10Y, bond, Ultra bond).

Exhibit 34: Decomposing a Treasury yield

Building block	Meaning	Where to observe it
Expected path of policy rates	Average expected fed funds rate over the bond's life	Fed funds and SOFR futures, OIS; dominates the 2-year
Term premium	Extra yield for bearing duration risk (inflation uncertainty, supply, volatility)	Model estimates: NY Fed ACM, Fed Board Kim-Wright; dominates the 10-30-year moves in fiscal-supply episodes
Real yield	Inflation-protected yield	TIPS yields (DFII5, DFII10)
Breakeven inflation	Nominal minus real yield = market inflation compensation	T5YIE, T10YIE, 5y5y forward (T5YIFR)

Source: compiled for this guide.

Curve shapes and spreads

- **2s10s** (10Y minus 2Y) and **3m10y** (10Y minus 3-month bill) measure the slope; the New York Fed's recession-probability model uses 3m10y.
- **Bull steepening** (front end falls faster) = markets price Fed cuts, often around recessions; **bear steepening** (long end rises faster) = term premium, inflation or fiscal concerns; **bear flattening** = hawkish Fed repricing; **bull flattening** = growth fears and falling long-term inflation expectations.
- The 2022-24 inversion was the longest on record; it un-inverted in 2024-25 without (so far) a recession, a reminder that the signal has long and variable lags.

Auctions and supply

Treasury sells bills weekly and coupons monthly. Watch the **tail** (auction yield vs pre-auction 'when-issued' yield), **bid-to-cover** ratio and the share taken by **indirect bidders** (a proxy for foreign official and investor demand). The Quarterly Refunding Announcement sets coupon sizes.

2. What It Implies & How It Affects Macro Strategy

The world's discount rate

The 10-year Treasury is the reference risk-free rate for global asset pricing: equity valuations (earnings yield vs real yield), mortgage rates (30-year mortgage ~ 10Y + spread), corporate bond yields (Treasury + credit spread) and emerging-market borrowing costs all key off it. Rising **real** yields tighten financial conditions and hit long-duration assets (growth stocks, gold historically, unprofitable tech); falling real yields do the reverse.

How strategists use it

- **Front end vs Fed:** the 2-year moves with expected policy; a 2-year yield well below fed funds signals the market expects cuts.
- **Data sensitivity:** on CPI and payroll days, 2-year yields often move 10-20bp; the long end responds more to inflation expectations, term premium and supply.
- **Regime reading:** stocks and yields both falling = growth scare; yields up with stocks down = inflation or fiscal scare; real yields vs breakevens show which is driving.
- **Stock-bond correlation:** positive in inflation-driven regimes (both fall together, as in 2022), negative in growth-driven regimes - it determines how well Treasuries hedge equity risk.
- **Cross-border:** US-Japan, US-Germany and US-Taiwan yield differentials drive FX and the hedging decisions of Asian life insurers, who are large holders of USD bonds.

3. How to Get the Data Free

Exhibit 35: Where to get Treasury yield data for free

Source	What you get	Notes
home.treasury.gov (Daily Treasury Par Yield Curve Rates)	Daily par, real (TIPS), bill and long-term rates since 1990	CSV/XML downloads
FRED	DGS1MO-DGS30, T10Y2Y, T10Y3M, DFII10, T10YIE, T5YIFR, THREEFYTP10	Daily, with API
treasurydirect.gov; fiscaldata.treasury.gov	Auction schedules, results (bid-to-cover, allotments)	Free
newyorkfed.org	ACM term premium; yield-curve recession probability	Excel downloads
CME Group	Delayed Treasury futures quotes, settlements, volume and open interest	See CME chapter

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 18 | MONETARY POLICY & RATES

CME Group Data & the FedWatch Tool

Bottom line

- **FedWatch** converts fed funds futures prices into market-implied probabilities for each FOMC meeting.
- CME publishes free settlement prices, volume and open interest for rates, equity, energy, metals and FX futures.
- Compare market pricing with the Fed's dot plot: the gap is where policy surprises come from.

At a glance

Indicator type	Market prices & derived probabilities
Lead / lag	Real time
Key question	What markets price for the Fed and for key assets, and how positioned they are
Publisher	CME Group (owner of CME, CBOT, NYMEX and COMEX derivatives exchanges)
Frequency	Real-time trading (delayed quotes free); daily settlement, volume and open interest; FedWatch probabilities update intraday
Headline	FedWatch: market-implied probabilities of each FOMC outcome; futures and options prices, volume, open interest
FRED codes	Not on FRED (CME data are proprietary; free access via cmegroup.com)
Taipei time	Nearly 24h electronic trading

1. Methodology

How FedWatch computes probabilities

30-Day Fed Funds futures (ticker ZQ) settle on the average daily effective fed funds rate over the contract month; the implied rate is 100 minus the futures price. For a month containing an FOMC meeting, the average rate is a weighted mix of the pre- and post-meeting rates (weighted by the days before and after the meeting). FedWatch backs out the expected post-meeting rate and expresses it as probabilities of 25bp moves, chaining meetings forward. Further out, **SOFR futures** (SR1, SR3) show the expected rate path for the next several years.

Exhibit 36: Key CME contracts for macro strategy

Asset class	Benchmark contracts	Macro use
Short-term rates	30-Day Fed Funds (ZQ), 3-Month SOFR (SR3)	Expected policy path, FedWatch, cut/hike pricing
Treasuries	2Y (ZT), 5Y (ZF), 10Y (ZN), Ultra 10Y (TN), Bond (ZB), Ultra Bond (UB)	Duration hedging, curve trades, basis trade; positioning via open interest
Equity indices	E-mini S&P 500 (ES), E-mini Nasdaq-100 (NQ), E-mini Russell 2000 (RTY), E-mini Dow (YM)	Overnight reaction to Asian/European news and data; options-implied event moves
Energy	WTI crude (CL), natural gas (NG), RBOB gasoline (RB), heating oil (HO)	Oil curve, crack spreads
Metals	COMEX gold (GC), silver (SI), copper (HG)	Safe haven, industrial cycle ('Dr Copper')
FX	Euro (6E), yen (6J), and other currency futures	Dollar exposure, positioning
Volatility	CVOL indices (implied volatility benchmarks across asset classes)	Cross-asset risk gauges

Source: CME Group.

Reading volume and open interest

- **Rising prices + rising open interest** = new positions supporting the trend; rising prices with falling open interest = short covering, often less durable.
- The CFTC's weekly **Commitments of Traders** report breaks CME open interest down by trader type (asset managers, leveraged funds, dealers) - a positioning gauge for crowded trades.
- Exchange margin changes can force deleveraging in volatile markets (e.g. metals in 2025).

2. What It Implies & How It Affects Macro Strategy

The market's view versus the Fed's

FedWatch is the quickest way to see what is priced for the next FOMC meeting and the year ahead. Comparing it with the Fed's dot plot and economists' forecasts shows where surprises can come from: if markets price three cuts while the dots show one, either the data must weaken or rates must reprice higher. Fed officials themselves watch market pricing and often avoid surprising it close to a meeting.

How strategists use it

- **Data-day attribution:** watch how the probability of a cut at the next meeting moves after CPI or payrolls - it quantifies how much a release changed the policy outlook.
- **Asymmetric opportunities:** when pricing is near 100% for an outcome, the risk is skewed to the surprise side.
- **Caveats:** futures prices embed risk premia and liquidity effects, so FedWatch probabilities are not pure forecasts; the tool assumes 25bp increments.
- **Overnight futures** (ES, NQ) and Treasury futures are how Asia-based investors trade US macro news outside the US cash session.

3. How to Get the Data Free

Exhibit 37: Where to get CME data for free

Source	What you get	Notes
cmegroup.com/markets/interest-rates/cme-fedwatch-tool	FedWatch probabilities, history, dot-plot comparison	Free (some features need a free login)
cmegroup.com (product pages)	Delayed quotes, daily settlements, volume and open interest, contract specs	Daily Bulletin PDFs for history
CVOL indices on cmegroup.com	Implied volatility indices by asset class	Free charts
cftc.gov	Commitments of Traders (weekly, Friday 3:30 p.m. ET)	Historical files free

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 19 | FINANCIAL MARKETS

US Equity Indices (incl. Nasdaq-100) & VIX

Bottom line

- Equities lead the economy and set financial conditions via the wealth effect.
- Ratios carry the macro signal: **SOX/S&P** for the tech/AI cycle, **Russell/S&P** for domestic rate sensitivity.
- VIX / sqrt(12) approximates the expected 1-month S&P 500 move; term-structure inversion signals stress.

At a glance

Indicator type	Market prices
Lead / lag	Leading (real time)
Key question	Risk appetite, financial conditions and the earnings outlook
Covered	S&P 500, Dow Jones Industrial Average, Nasdaq Composite, PHLX Semiconductor Index (SOX), Russell 2000, Cboe Volatility Index (VIX)
Nature	Real-time market prices that both reflect and drive macro conditions (wealth effect, financial conditions, confidence)
FRED codes	SP500, DJIA, NASDAQCOM, VIXCLS (FRED shows only ~10 years for some licensed index series)
Taipei time	NYSE/Nasdaq cash session 22:30-05:00 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

Exhibit 38: US equity indices and VIX: how they are built

Index	Constituents	Weighting	Key methodology points
S&P 500	~500 large-cap US companies (~80% of US market cap)	Float-adjusted market cap	Selected by a committee at S&P Dow Jones Indices. Eligibility: US domicile, minimum market-cap threshold (raised periodically; in the low-$20bn range in 2025), liquidity, float, and positive GAAP earnings (sum of last four quarters and the latest quarter). Quarterly rebalances; ad hoc additions/deletions
Dow Jones Industrial Average	30 blue-chip US companies	**Price-weighted**	Sum of share prices divided by the 'Dow divisor' (adjusted for splits and changes). A $1 move in any stock moves the index equally, so high-priced stocks dominate regardless of size. Committee-selected; excludes transport and utilities (separate averages). Oldest index (1896)
Nasdaq Composite	All (~3,000+) common stocks listed on the Nasdaq exchange, incl. foreign issuers	Market cap	Heavily tech and growth tilted; top mega-caps dominate. Not the same as the **Nasdaq-100** (100 largest non-financials, modified cap weight, the basis for QQQ)
Nasdaq-100	100 largest non-financial companies listed on Nasdaq	Modified market cap	Caps reduce concentration in mega-caps (special rebalances when the largest names grow too big); quarterly rebalance and annual December reconstitution. Basis for QQQ and the E-mini Nasdaq-100 futures (NQ)
PHLX Semiconductor (SOX)	30 largest US-listed semiconductor companies (design, manufacturing, equipment, distribution), incl. ADRs like TSMC and ASML	Modified market cap	Caps limit concentration (the largest names capped at around 8-12%, the rest at ~4%) with quarterly rebalancing. Owned by Nasdaq; tracked by SOXX and leveraged ETFs
Russell 2000	Smallest ~2,000 companies of the Russell 3000 (US small caps)	Float-adjusted market cap	Rules-based (no committee): rank by market cap on the rank day. Historically reconstituted annually in June; FTSE Russell has moved to a semi-annual reconstitution (June and December) starting 2026. ~40% of constituents are unprofitable; higher share of floating-rate debt

Index	Constituents	Weighting	Key methodology points
VIX	Not a stock index: derived from S&P 500 (SPX) options	-	Model-free estimate of 30-day expected volatility of the S&P 500, computed from out-of-the-money SPX puts and calls across a wide strike range with 23-37 days to expiry, interpolated to 30 days. Expressed as annualised % (variance-swap style). Calculated real time by Cboe

Source: S&P Dow Jones Indices, Nasdaq, FTSE Russell, Cboe.

Reading the VIX

- VIX / sqrt(12) approximates the expected 1-month move: VIX 16 implies roughly +/-4.6% over a month (one standard deviation); VIX 32 implies ~9%.
- Typical regimes: ~12-15 calm, ~20 elevated, 30+ stress, 40-80 crisis (2008 peak ~80; March 2020 ~83 close; intraday spikes like August 2024).
- VIX is mean-reverting and usually rises when stocks fall (negative spot-vol correlation), because investors bid for downside puts.
- **Term structure:** VIX9D, VIX (30d), VIX3M and VIX futures. Normally in contango (longer > shorter); **inversion** (spot above 3-month) signals acute stress and has often marked tactical capitulation points.
- Related gauges: MOVE (Treasury volatility, ICE BofA), VVIX (vol of VIX), SKEW, VXN (Nasdaq-100), RVX (Russell 2000).

2. What It Implies & How It Affects Macro Strategy

Equities as macro indicators

- **Leading indicator:** stock prices discount future earnings and so lead the economy; the S&P 500 is a component of the Conference Board Leading Economic Index. But 'the market has predicted nine of the last five recessions' - drawdowns without recessions are common.
- **Wealth effect and financial conditions:** higher equity prices boost spending (especially among higher-income households who now drive a large share of consumption) and loosen financial conditions; the Fed watches indices such as the Chicago Fed NFCI, which includes equity volatility.
- **Valuation vs rates:** compare the forward earnings yield with real 10-year yields (equity risk premium); a thin ERP makes equities vulnerable to rising real rates.

What each index tells a macro strategist

Exhibit 39: What each index tells a macro strategist

Index / ratio	Macro reading
S&P 500	Broad corporate earnings expectations and global risk appetite; dominated by mega-cap tech (top 10 > ~35% of weight), so also check the **equal-weight S&P 500** (RSP) for breadth
Dow Jones	Old-economy, value and cyclical tilt (industrials, financials, healthcare); headline familiarity matters for retail sentiment; price weighting distorts
Nasdaq Composite	Long-duration growth: most sensitive to real yields and AI/tech capex narratives; outperforms when real rates fall
SOX	The global tech and electronics cycle, AI capex (data centres) and Asian export economies (Taiwan, Korea). SOX tends to lead ISM New Orders and global trade; SOX/S&P 500 ratio is a high-beta cycle gauge. Sensitive to export controls and tariffs
Russell 2000	Domestic US economy and credit conditions: more floating-rate debt, less pricing power, more regional banks. Outperforms on rate cuts with stable growth (early cycle); underperforms on higher-for-longer. Russell 2000/S&P 500 ratio = domestic cyclical and rates sentiment
VIX	Risk-off barometer; drives volatility-targeting and risk-parity deleveraging. Spikes tighten financial conditions and can trigger 'Fed put' speculation; persistently low VIX encourages leverage and short-vol positioning (fragility)

Source: compiled for this guide.

Strategy uses

- **Regime identification:** stocks up, yields up = growth optimism; stocks down, yields up = inflation/policy scare (hawkish); stocks down, yields down = growth scare; stocks up, yields down = Goldilocks/disinflation.
- **Cyclical vs defensive** and **small vs large** ratios confirm or contradict macro data (e.g. Russell 2000 outperformance alongside rising ISM confirms recovery).
- **Event pricing:** SPX options imply the expected move on CPI, payrolls and FOMC days; compare realised to implied to judge whether data is being over- or under-priced.
- **Taiwan and Asia link:** SOX is the key external driver of TAIEX and KOSPI; US semiconductor moves overnight often set Asia's open.

3. How to Get the Data Free

Exhibit 40: Where to get equity and volatility data for free

Source	What you get	Notes
FRED	SP500, DJIA, NASDAQCOM, VIXCLS (daily closes)	S&P/Dow series limited to ~10 years by licence
Cboe (cboe.com)	Full VIX history since 1990 (CSV download), VIX term structure, VIX9D/VIX3M, methodology white paper	cboe.com/tradable_products/vix
Yahoo Finance / Stooq	Daily OHLC history for ^GSPC, ^DJI, ^IXIC, ^SOX, ^RUT, ^VIX; CSV downloads; yfinance Python library	Unofficial but widely used
Index providers	S&P DJI (spglobal.com/spdji), Nasdaq (nasdaq.com/market-activity/index/sox), FTSE Russell - methodology documents, fact sheets, constituents	Methodology PDFs free
Valuation	Robert Shiller's data (CAPE, earnings, dividends since 1871); S&P DJI earnings spreadsheet; multpl.com	Free
Financial conditions	Chicago Fed NFCI (FRED: NFCI)	Weekly

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 20 | FINANCIAL MARKETS

DXY: US Dollar Index (and the Fed Broad Dollar Index)

Bottom line

- **DXY** is the dollar against six developed-market currencies, dominated by the euro (~58%).
- The Fed's **Broad Dollar Index** is the better macro measure (26 currencies incl. China, Mexico, Taiwan).
- A strong dollar tightens global financial conditions and pressures EM assets and commodity prices.

At a glance

Indicator type	Market price
Lead / lag	Real time
Key question	Global dollar strength, US vs world rate and growth differentials
Publisher	ICE Futures U.S. (DXY); Federal Reserve Board (Broad Dollar Index, H.10 release)
Frequency	DXY real time, nearly 24 hours; Fed indices daily, published weekly on the H.10 release
Headline	DXY index level (March 1973 = 100); Nominal Broad Dollar Index (January 2006 = 100)
FRED codes	DTWEXBGS (broad), DTWEXAFEGS (vs advanced foreign economies), DTWEXEMEGS (vs emerging markets), DEXTAUS (TWD per USD)
Taipei time	Trades nearly 24h

1. Methodology

DXY is a **geometric weighted average** of the dollar against six currencies. Its weights were set when the euro replaced its predecessor currencies in 1999 and have not changed since, so it does not reflect today's US trade partners (no Chinese yuan, Mexican peso, Korean won or Taiwan dollar).

Exhibit 41: DXY composition vs the Fed Broad Dollar Index

Feature	ICE DXY	Fed Nominal Broad Dollar Index
Currencies	EUR 57.6%, JPY 13.6%, GBP 11.9%, CAD 9.1%, SEK 4.2%, CHF 3.6%	26 currencies incl. CNY, MXN, CAD, EUR, JPY, KRW, TWD, INR, VND
Weights	Fixed since 1999	Trade-weighted, updated annually (goods and services trade)
Strength	Traded (futures and options); the market's headline dollar gauge	Better measure of the dollar's macro impact on US trade and inflation
Sub-indices	-	Advanced foreign economies and emerging-market economies indices; real (inflation-adjusted) versions

Source: ICE, Federal Reserve Board.

2. What It Implies & How It Affects Macro Strategy

What moves the dollar

- **Rate and growth differentials:** US yields and growth relative to the rest of the world are the main medium-term drivers.
- **The 'dollar smile':** the dollar tends to rise both in global risk-off episodes (safe-haven demand) and when US growth outperforms, and to fall when global growth is broad and synchronised.
- **Capital flows and hedging:** foreign demand for US assets (which finances the current account deficit) and hedge ratios of foreign investors; rising hedging by Asian and European investors weighed on the dollar in 2025.
- **Policy and credibility:** trade policy, fiscal deficits and concerns about Fed independence; the dollar fell about 10% in the first half of 2025, its weakest first half in decades.

How strategists use it

- **Global financial conditions:** a strong dollar tightens conditions for EM borrowers with dollar debt and tends to pressure EM equities and commodities (priced in dollars).
- **US earnings:** roughly 40% of S&P 500 revenue comes from abroad, so a weaker dollar boosts reported earnings (especially for tech and materials).
- **Inflation:** a weaker dollar raises import prices and core goods inflation with a lag; a stronger dollar is disinflationary.
- **Taiwan angle:** USD/TWD moves affect exporters' margins, foreign equity flows into TAIEX and the large FX exposure of Taiwanese life insurers; sharp TWD appreciation (as in May 2025) can force hedging flows.
- **Asset allocation:** USD weakness tends to favour non-US equities, EM local bonds and gold.

3. How to Get the Data Free

Exhibit 42: Where to get dollar data for free

Source	What you get	Notes
theice.com	DXY methodology, delayed futures quotes	Index rules document free
federalreserve.gov/releases/h10	Broad, AFE, EME indices (nominal and real), bilateral exchange rates, weights	Weekly release, daily data
FRED	DTWEXBGS, DTWEXAFEGS, DTWEXEMEGS, DEXJPUS, DEXUSEU, DEXTAUS	Daily
Yahoo Finance	DX-Y.NYB daily history	Unofficial
Central Bank of the Republic of China (Taiwan)	TWD exchange rates and FX statistics	cbc.gov.tw

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 21 | FINANCIAL MARKETS

Gold Price

Bottom line

- Benchmark is the **LBMA Gold Price** (twice-daily London auction); COMEX futures dominate trading.
- Classic driver is real yields, but since 2022 **central-bank buying** has broken that relationship.
- Gold now trades as a hedge against geopolitical, sanctions and fiscal/dollar-debasement risk.

At a glance

Indicator type	Market price
Lead / lag	Real time
Key question	Real-yield, safe-haven and reserve-diversification demand
Benchmarks	LBMA Gold Price (administered by ICE Benchmark Administration); COMEX gold futures (CME); spot XAU/USD
Frequency	LBMA auctions twice daily at 10:30 and 15:00 London time; futures and spot trade nearly 24 hours; World Gold Council data quarterly (Gold Demand Trends) and monthly (ETF flows, central banks)
Headline	US dollars per troy ounce
FRED codes	Not available (LBMA prices were removed from FRED in 2022); use LBMA, CME, World Gold Council
Taipei time	LBMA auctions 18:30 and 23:00 (one hour earlier during UK summer time); futures nearly 24h

1. Methodology

The **LBMA Gold Price** is set in an electronic, physically settled auction in London at 10:30 and 15:00, where participating banks match buy and sell interest until the market clears. It is the reference price for contracts, central banks and ETFs. Price discovery during the trading day happens mainly in COMEX **futures** (GC, 100 troy ounces) and the London over-the-counter spot market. Shanghai Gold Exchange prices reflect Chinese demand; a large Shanghai premium signals strong Asian buying.

Exhibit 43: Gold demand and supply

Component	Notes
Jewellery	Largest demand category historically; price-sensitive (India, China)
Investment: bars, coins, ETFs	ETF flows (e.g. GLD) respond to real yields and momentum; bar and coin demand to fear and local currency weakness
Central banks	Net buying above 1,000 tonnes a year in 2022-24 as reserve managers diversified away from the dollar after Russian reserves were frozen
Technology	Electronics; small share
Supply	Mine production (~3,600 t/yr) plus recycling; supply responds slowly to price

Source: World Gold Council.

2. What It Implies & How It Affects Macro Strategy

What drives gold

- **Real yields:** gold pays no income, so the opportunity cost is the real interest rate. For two decades gold moved inversely to 10-year TIPS yields.
- **The regime change since 2022:** real yields rose sharply yet gold surged to records - above $4,000/oz in 2025 - because central-bank and Asian demand overwhelmed the real-yield signal.
- **The dollar:** a weaker dollar usually supports gold prices.
- **Geopolitics, sanctions and fiscal risk:** gold is the classic hedge against war, sanctions risk and concerns about fiscal sustainability or currency debasement ('debasement trade').

How strategists use it

- **Portfolio hedge:** gold has tended to hold up when both equities and bonds fall (inflationary or credibility shocks), making it a diversifier when the stock-bond correlation is positive.
- **Signal of trust in fiat and US assets:** gold outperforming Treasuries during a period of US policy uncertainty is read as a vote on institutional credibility.
- **Gold vs copper:** the copper/gold ratio tracks global growth expectations and correlates with the 10-year yield.
- **Watch positioning:** CFTC managed-money positions and ETF holdings show how crowded the trade is; sharp drawdowns often follow extreme positioning.

3. How to Get the Data Free

Exhibit 44: Where to get gold data for free

Source	What you get	Notes
lbma.org.uk	LBMA precious metals prices and methodology	Historical prices accessible on the site
gold.org (Goldhub)	Gold Demand Trends, central-bank purchases, ETF holdings and flows, price data in many currencies	Free registration for downloads
cmegroup.com	Delayed COMEX quotes, settlements, volume, open interest, warehouse stocks	Free
Yahoo Finance / Stooq	GC=F futures and XAUUSD history	Unofficial
IMF IFS / COFER	Official gold reserves by country	See IMF chapter

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 22 | FINANCIAL MARKETS

Bitcoin

Bottom line

- A fixed-supply (21 million) digital asset trading 24/7; US **spot ETFs** since 2024 made flows a key driver.
- Mostly behaves as a **high-beta, liquidity-sensitive** risk asset tied to real yields, the dollar and the Nasdaq.
- Watch ETF flows, CME basis, funding rates and open interest to judge whether moves are spot- or leverage-driven.

At a glance

Indicator type	Market price (digital asset)
Lead / lag	Real time (24/7)
Key question	Speculative risk appetite and global liquidity; adoption via ETFs and policy
Benchmarks	Spot prices on major exchanges (e.g. Coinbase, Binance, Kraken); CME CF Bitcoin Reference Rate (BRR), used to settle CME futures; CME Bitcoin and Micro Bitcoin futures and options; US spot bitcoin ETFs
Frequency	Trades 24 hours a day, 7 days a week; BRR fixed daily at 4:00 p.m. London; ETF flows reported daily
Headline	BTC/USD price; market capitalisation; ETF net flows; futures basis, funding rates and open interest
FRED codes	Not a core FRED series; use exchange data, CME, ETF issuers and free crypto data providers
Taipei time	Trades 24/7; CME CF reference rate fixed at 00:00 (one hour earlier during UK summer time)

1. Methodology

How the asset works

Bitcoin is a decentralised digital asset launched in 2009. Transactions are recorded on a public blockchain secured by **proof-of-work** mining: miners compete to add blocks roughly every 10 minutes and receive newly issued bitcoin plus fees. Supply is capped at **21 million** coins, and the block reward is cut in half every 210,000 blocks (about every four years) - the **halving**. The most recent halving was in April 2024; the next is expected around 2028. More than 19.5 million coins have already been mined, so new supply is now small relative to the existing stock.

Exhibit 45: Key market structures and data

Element	What it is	What it tells you
Spot exchanges	Centralised venues trading BTC against USD and stablecoins (USDT, USDC)	Price discovery; the Coinbase premium (US vs offshore price) signals US demand
US spot bitcoin ETFs	Exchange-traded funds holding physical bitcoin, approved in January 2024	Daily net flows measure institutional and wealth-management demand
CME futures and options	Regulated, cash-settled against the BRR; Bitcoin (5 BTC) and Micro (0.1 BTC) contracts	Basis (futures premium over spot) shows leveraged long demand; CFTC COT shows positioning
Perpetual swaps (offshore)	Futures with no expiry; longs and shorts exchange **funding** payments	Positive, rising funding = crowded leveraged longs; liquidation cascades amplify moves
On-chain data	Hash rate, active addresses, exchange inflows/outflows, holder cohorts	Network security, usage, and whether long-term holders are distributing
Stablecoin supply	Total issuance of dollar stablecoins	A proxy for liquidity available to crypto markets

Source: compiled for this guide.

Policy and structural milestones

- Spot ETF approval (January 2024) opened bitcoin to mainstream portfolios and made ETF flows a key driver.
- In 2025 the US government established a Strategic Bitcoin Reserve by executive order and enacted a federal stablecoin law (the GENIUS Act), shifting US policy towards integration rather than restriction.
- Corporate treasury buyers (listed companies holding bitcoin on their balance sheets) became a significant source of demand.

- Bitcoin set record highs above $120,000 in 2025 before a sharp, leverage-driven sell-off in October 2025 - a reminder of its volatility (historically several times that of the S&P 500).

2. What It Implies & How It Affects Macro Strategy

A liquidity-sensitive risk asset

Despite the 'digital gold' narrative, bitcoin has mostly traded as a **high-beta risk asset**. Its price has correlated with global liquidity (central-bank balance sheets, broad money growth, stablecoin supply), falling real yields, a weaker dollar and the Nasdaq. In 2022 it fell about 75% from peak as the Fed hiked aggressively. At times of banking stress or debasement concerns, it has also behaved more like gold. Because it trades around the clock, it is often the first market to react to weekend news.

How strategists use it

- **Liquidity and risk-appetite gauge:** bitcoin breaking out or breaking down alongside high-beta tech can confirm a shift in speculative risk appetite; divergence from equities is worth investigating.
- **Rates sensitivity:** dovish Fed repricing and falling real yields have tended to support bitcoin; hawkish surprises hit it hard, often more than equities.
- **Flows and leverage:** ETF flows, CME basis, perpetual funding rates and open interest show whether a rally is driven by spot demand or by leverage that can unwind quickly.
- **Cross-asset contagion:** crypto-exposed equities (miners, exchanges, corporate holders) and stablecoin reserves (largely held in Treasury bills) link crypto to traditional markets; stablecoin growth adds to T-bill demand.
- **Portfolio role:** as a small diversifier, its value depends on correlation regime; size positions for very high volatility and drawdowns.
- **Taiwan angle:** local trading is regulated by Taiwan's Financial Supervisory Commission through registered virtual asset service providers; US spot ETFs are accessible via overseas brokers and sub-brokerage.

3. How to Get the Data Free

Exhibit 46: Where to get bitcoin data for free

Source	What you get	Notes
CoinGecko; CoinMarketCap	Prices, market caps, volumes, historical data across exchanges	Free web and API tiers
cmegroup.com; cfbenchmarks.com	CME futures and options settlements, volume, open interest; BRR methodology	Delayed data free
ETF issuer websites; Farside Investors	Daily holdings and net flows of US spot bitcoin ETFs	Free
Coin Metrics Community; Glassnode (free tier); mempool.space	On-chain data: hash rate, addresses, supply, fees	Free tiers with limits
cftc.gov	Commitments of Traders for CME bitcoin futures	Weekly
Yahoo Finance	BTC-USD daily history	Unofficial

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 23 | ENERGY & COMMODITIES

WTI and Brent Crude Oil

Bottom line

- **WTI** prices US inland crude at Cushing; **Brent** prices global seaborne crude.
- Oil drives headline inflation and short-term household inflation expectations; the Fed looks through supply shocks.
- Supply-driven spikes are stagflationary; demand-driven rallies are benign. Time spreads show tightness.

At a glance

Indicator type	Market price
Lead / lag	Real time
Key question	Energy inflation, terms of trade and geopolitical risk
What they are	The two global benchmark crude oil prices, quoted in USD per barrel; real-time market prices rather than statistical releases
Primary venues	**WTI**: NYMEX/CME light sweet crude futures (ticker CL). **Brent**: ICE Brent futures (ticker B / CO)
Key data events	API inventories (Tue 16:30 ET), **EIA Weekly Petroleum Status Report** (Wed 10:30 ET), OPEC+ meetings, Baker Hughes rig count (Fri), CFTC Commitments of Traders (Fri), monthly EIA STEO / OPEC MOMR / IEA OMR
FRED codes	DCOILWTICO (WTI spot, daily), DCOILBRENTEU (Brent spot, daily)
Taipei time	Futures nearly 24h

1. Methodology

West Texas Intermediate (WTI)

A light (API gravity ~39-40), sweet (low-sulphur, ~0.3-0.4%) US crude. The benchmark NYMEX futures contract is 1,000 barrels with **physical delivery at Cushing, Oklahoma**, a landlocked pipeline and storage hub. WTI therefore reflects US mid-continent supply and demand, Cushing storage levels and pipeline/export capacity. The April 2020 negative price episode occurred because expiring contracts required physical delivery while Cushing storage was nearly full.

Brent

Originally North Sea Brent field crude, now a basket: Brent, Forties, Oseberg, Ekofisk, Troll (BFOET) and, since 2023, US **WTI Midland** delivered to Rotterdam, which is frequently the cheapest grade setting the price. ICE Brent futures are **cash-settled** against the ICE Brent Index, tied to the physical 'Dated Brent' assessment by Platts. Because it is waterborne, Brent prices roughly two-thirds of internationally traded crude and reflects global seaborne balances and geopolitical risk (Middle East, Russia, shipping chokepoints).

Structure and spreads

Exhibit 47: Oil market structure and spreads

Concept	Meaning	Signal
Brent-WTI spread	Brent minus WTI (usually +$3-5)	Wider = US export/transport constraints or international tightness; narrow = US export arbitrage working
Backwardation	Front-month > later months	Tight market, low inventories; positive roll yield for long futures holders
Contango	Front-month < later months	Oversupply, inventory building; storage economics; negative roll
Crack spreads	Refined product (gasoline, diesel) minus crude	Refining margins; diesel cracks track industrial/freight demand
Time spreads (e.g. M1-M6)	Calendar spread	Cleaner read on physical tightness than flat price

Source: CME Group, ICE, compiled for this guide.

Fundamental drivers

- **Supply:** OPEC+ quotas and spare capacity (concentrated in Saudi Arabia/UAE), US shale output and productivity (rig counts, breakevens around $55-65 for new wells), Russia sanctions, Iran and Venezuela exports.
- **Demand:** global GDP, China (imports, stockpiling, EV adoption), jet fuel, petrochemicals; OECD demand is structurally flat.
- **Inventories:** OECD commercial stocks vs 5-year average, US crude and products (EIA weekly), the Strategic Petroleum Reserve.
- **Financial:** USD strength (oil priced in dollars), speculative positioning (CFTC managed money), volatility and risk appetite.

2. What It Implies & How It Affects Macro Strategy

Inflation channel

Energy is ~6-7% of the CPI basket, but gasoline is highly visible: a $10/bbl change in crude typically moves retail gasoline by roughly 25 cents/gallon and headline CPI by a few tenths over a few months. Oil spikes push up **headline** inflation and short-term **household inflation expectations** (Michigan 1-yr is highly correlated with gasoline prices). Central banks generally 'look through' supply-driven energy shocks to core inflation unless expectations de-anchor or second-round effects appear (airfares, transport, goods).

Growth and terms of trade

- For consumers, higher oil acts like a tax that reduces real disposable income. Oil price spikes preceded many post-war US recessions (1973, 1979, 1990, 2008).
- Since the shale revolution the US is a net petroleum exporter, so the GDP effect of higher oil is more balanced (energy capex and producer income offset consumer drag), but distributional effects remain.
- For importers (Japan, India, Korea, Taiwan, Eurozone) higher oil worsens the terms of trade and current account, weakening currencies; exporters (Canada, Norway, Saudi Arabia, Brazil) benefit.

Cross-asset uses

- **Stagflation risk:** a supply-driven spike (oil up, equities down, breakevens up) is the worst mix for balanced portfolios; a demand-driven rise (oil up with copper and equities) is benign.
- **Breakevens:** oil explains a large share of short-dated TIPS breakeven moves.
- **Credit:** energy is a large sector of the US high-yield market; oil collapses (2015-16, 2020) widen HY spreads.
- **Equities:** energy sector earnings, airlines and transport costs; the oil-equity correlation flips by regime.
- **FX:** CAD, NOK, MXN and petro-EM currencies; oil strength supports them versus JPY, INR, KRW, TWD.
- **Geopolitical hedging:** long oil or long energy equities can hedge Middle East escalation risk.

3. How to Get the Data Free

Exhibit 48: Where to get oil data for free

Source	What you get	Notes
EIA (eia.gov/petroleum)	Daily WTI and Brent spot history (since 1986/87), Weekly Petroleum Status Report, Short-Term Energy Outlook, gasoline prices	EIA Open Data API with free key
FRED	DCOILWTICO, DCOILBRENTEU, GASREGW (retail gasoline)	Daily; sourced from EIA
CME Group / ICE websites	Delayed futures quotes, settlements and full futures curve	Free, delayed ~10-20 min
CFTC (cftc.gov)	Commitments of Traders positioning, weekly	Free historical files
OPEC MOMR, IEA OMR highlights	Monthly supply/demand balances	OPEC report free; IEA highlights free, full report paid
Baker Hughes (rigcount.bakerhughes.com)	Weekly US and international rig counts	Free

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 24 | ENERGY & COMMODITIES

EIA: US Energy Information Administration Reports

Bottom line

- The **Weekly Petroleum Status Report** (Wednesday) is the key weekly physical data point for oil.
- Natural Gas Storage (Thursday) and the monthly **STEO** forecast round out the core EIA releases.
- Inventories vs the five-year average and Cushing stocks explain moves in prices and time spreads.

At a glance

Indicator type	Official energy statistics
Lead / lag	Weekly / monthly
Key question	Physical oil and gas balances behind energy prices
Publisher	US Energy Information Administration, statistical agency of the US Department of Energy
Frequency	Weekly Petroleum Status Report (Wednesday 10:30 ET; Thursday after holidays); Weekly Natural Gas Storage Report (Thursday 10:30 ET); retail gasoline and diesel prices (Monday); Short-Term Energy Outlook (STEO, monthly); Petroleum Supply Monthly; Annual Energy Outlook
Headline	Weekly changes in crude oil, gasoline and distillate inventories (million barrels); natural gas storage change (billion cubic feet)
FRED codes	WCESTUS1 (commercial crude stocks), GASREGW (retail gasoline), DHHNGSP (Henry Hub gas price)
Taipei time	Petroleum report Wed 23:30; gas storage Thu 23:30 (one hour earlier while US daylight saving time is in force, Mar-Nov)

1. Methodology

Exhibit 49: Core EIA releases for markets

Release	Key contents	Notes
Weekly Petroleum Status Report (WPSR)	Crude inventories (ex-SPR), **Cushing** stocks, gasoline and distillate inventories, refinery runs and utilisation, imports/exports, product supplied (implied demand), estimated production	Industry group API publishes its own survey on Tuesday 4:30 p.m. ET as a preview
Weekly Natural Gas Storage Report	Net injection/withdrawal vs consensus and five-year average; regional storage	Key driver of Henry Hub natural gas futures
Gasoline & diesel retail prices	National and regional pump prices	Nowcasts the CPI energy component
Short-Term Energy Outlook (STEO)	18-24 month forecasts of global oil balances, US production, prices, power demand	Includes US shale productivity analysis
Petroleum Supply Monthly	Survey-based monthly production and demand	More accurate than weekly estimates; ~2-month lag
Annual Energy Outlook; electricity data	Long-term projections; power generation and demand	Data-centre electricity demand is a growing theme

Source: EIA.

2. What It Implies & How It Affects Macro Strategy

Physical data behind energy prices

Weekly inventory changes show whether the US market is tightening or loosening. Traders compare the reported change with consensus and with seasonal norms; draws larger than expected are bullish for crude and time spreads, builds bearish. **Cushing** stocks (the WTI delivery point) matter disproportionately for the front of the WTI curve. Refinery utilisation and product inventories drive crack spreads and pump prices.

How strategists use it

- **Oil trading:** the Wednesday report is the main scheduled weekly event for WTI; combine it with OPEC+ decisions and global balances for the medium-term view (see the WTI/Brent chapter).
- **Inflation nowcasting:** weekly retail gasoline prices give a near-real-time read on the CPI energy component and on household inflation expectations.
- **US production trend:** EIA's monthly data and STEO show whether shale output is growing or plateauing - important for the supply side of oil prices and for energy-sector equities and HY credit.
- **Natural gas and power:** storage levels, LNG exports and AI-driven power demand shape utilities' outlook and electricity inflation.

3. How to Get the Data Free

Exhibit 50: Where to get EIA data for free

Source	What you get	Notes
eia.gov/petroleum/supply/weekly	Weekly Petroleum Status Report tables and history	Release schedule on site
eia.gov/naturalgas/storage	Weekly storage report and history	Thursday
eia.gov/outlooks/steo	Short-Term Energy Outlook, data browser	Monthly
EIA Open Data API v2	Programmatic access to almost all EIA series	Free key; Excel add-in available
FRED	Selected EIA series (spot prices, gasoline, inventories)	Daily/weekly

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 25 | INTERNATIONAL INSTITUTIONS

IMF Data

Bottom line

- The **World Economic Outlook** (Apr/Oct) provides consistent growth and inflation forecasts for ~190 economies.
- The Fiscal Monitor, Global Financial Stability Report and US **Article IV** review frame structural risks.
- **COFER** shows the dollar's share of global FX reserves - the core data in the 'de-dollarisation' debate.

At a glance

Indicator type	International institution data
Lead / lag	Forecasts & structural data
Key question	Global and cross-country context: growth, fiscal, financial stability, reserves
Publisher	International Monetary Fund (Washington, DC)
Frequency	World Economic Outlook (WEO) in April and October with updates in January and July; Fiscal Monitor and Global Financial Stability Report (GFSR) in April and October; annual Article IV consultation with the US; COFER quarterly
Headline	Global and country forecasts for GDP, inflation, current account, fiscal balance and debt
FRED codes	Some IMF series are mirrored on FRED; primary source data.imf.org
Taipei time	Scheduled publications (WEO in April and October)

1. Methodology

Exhibit 51: Main IMF publications and datasets

Product	What it contains	Why it matters
World Economic Outlook (WEO) + database	Forecasts and history for ~190 economies: real GDP, inflation, unemployment, current account, government balance and debt	Consistent cross-country baseline; widely quoted headline global growth forecast
Fiscal Monitor	General-government deficits, debt and fiscal risks	Comparable fiscal data (general government, not just federal)
Global Financial Stability Report (GFSR)	Risks in markets, banks, non-bank finance	Flags vulnerabilities (leverage, private credit, asset valuations)
External Sector Report	Current account and exchange-rate assessments vs fundamentals	Estimates of over/under-valued currencies, including the dollar
Article IV (United States)	Annual review of US policies with staff recommendations	Independent view on US fiscal path, Fed policy, trade
COFER	Currency composition of official foreign-exchange reserves	Dollar share of reserves (~57-58% in recent data, down from ~70% in 2000)
International Financial Statistics (IFS) and other datasets	Exchange rates, interest rates, prices, balance of payments, reserves for all members	Standardised definitions

Source: International Monetary Fund.

2. What It Implies & How It Affects Macro Strategy

The global context for US macro

- **Relative growth:** WEO forecasts show whether the US is outgrowing the rest of the world - a key driver of the dollar and of US vs non-US equity allocation.
- **Forecast revisions** matter more than levels: successive downgrades signal a deteriorating global outlook (e.g. revisions after major tariff announcements in 2025).
- **Structural risks:** the Fiscal Monitor and GFSR quantify debt and financial-stability risks that can surface as term-premium or credit shocks.

- **Reserve currency status:** COFER is the core dataset in the de-dollarisation debate; changes are slow, and valuation effects (exchange-rate moves) distort quarterly shares.
- **Caveat:** IMF forecasts are produced at fixed points with policy assumptions and tend to lag turning points; use them as a consistent baseline, not a market signal.

3. How to Get the Data Free

Exhibit 52: Where to get IMF data for free

Source	What you get	Notes
imf.org/en/Publications/WEO	WEO reports and the full WEO database (Excel/CSV downloads)	Free
data.imf.org	IMF data portal: IFS, BOP, COFER, Direction of Trade, government finance	Free with API (SDMX)
imf.org/external/datamapper	Interactive charts of key WEO and Fiscal Monitor indicators	Quick comparisons
imf.org (Countries > United States)	Article IV reports and concluding statements	Annual

Source: publisher websites, compiled for this guide (access terms as of November 2026).

CHAPTER 26 | INTERNATIONAL INSTITUTIONS

World Bank Data

Bottom line

- **Global Economic Prospects** (Jan/Jun) is the World Bank's flagship global and EM growth forecast.
- **World Development Indicators** offer free long-run cross-country data on almost everything.
- The monthly **Pink Sheet** gives commodity price history back to 1960 - ideal for long-horizon analysis.

At a glance

Indicator type	International institution data
Lead / lag	Forecasts & long-run data
Key question	Global and EM growth outlook, long-run development and commodity price history
Publisher	World Bank Group (Washington, DC)
Frequency	Global Economic Prospects (GEP) in January and June; Commodity Markets 'Pink Sheet' monthly (early month); Commodity Markets Outlook in April and October; World Development Indicators updated periodically
Headline	Global, advanced-economy and EM/developing-economy growth forecasts; commodity prices and indices
FRED codes	Some World Bank series are mirrored on FRED; primary source data.worldbank.org
Taipei time	Scheduled publications (GEP in January and June)

1. Methodology

Exhibit 53: Main World Bank publications and datasets

Product	What it contains	Why it matters
Global Economic Prospects (GEP)	Growth forecasts for the world, regions and individual EMDEs; analysis of risks (trade, debt, commodity prices)	The other major institutional global forecast alongside the IMF WEO, with deeper EM focus; uses market exchange rates for aggregation
World Development Indicators (WDI)	1,400+ indicators for ~200 economies, often back to 1960: GDP, population, trade, energy, education, poverty	Free long-run cross-country data for structural analysis
Commodity Markets 'Pink Sheet'	Monthly prices of ~70 commodities (energy, metals, agriculture, fertilisers) and price indices since 1960	Long history for commodity cycles and inflation analysis
Commodity Markets Outlook	Twice-yearly forecasts for commodity prices	Context for oil and food inflation
International Debt Statistics; WITS	External debt of developing countries; trade and tariff data	EM debt vulnerability, tariff analysis

Source: World Bank.

2. What It Implies & How It Affects Macro Strategy

Why a US macro strategist uses it

- **Global demand for US exports and earnings:** EM growth forecasts (China, India, ASEAN) shape demand for US goods, commodities and S&P 500 overseas revenue.
- **Commodity cycles:** the Pink Sheet's 60+ years of data put current oil, copper, gold and food prices into historical context (real price levels, super-cycles).
- **EM risk:** debt statistics and growth downgrades flag countries vulnerable to a strong dollar and high US rates.
- **Structural themes:** demographics, productivity and energy transition data from WDI underpin long-horizon views (e.g. potential growth, labour-force trends).
- **Caveat:** like the IMF, World Bank forecasts are slow to change; the value is in consistency and breadth.

3. How to Get the Data Free

Exhibit 54: Where to get World Bank data for free

Source	What you get	Notes
worldbank.org/en/publication/global-economic-prospects	GEP reports and forecast tables	Free
data.worldbank.org; databank.worldbank.org	WDI and 80+ other databases; custom queries and downloads	Free
worldbank.org/en/research/commodity-markets	Pink Sheet monthly and annual price data (Excel)	Free
World Bank API (api.worldbank.org)	Programmatic access to WDI and more	No key required; Python: wbgapi

Source: publisher websites, compiled for this guide (access terms as of November 2026).

SYNTHESIS

Putting It Together: An Integrated Macro Framework

Growth x inflation regimes

Most macro allocation frameworks place the economy in one of four regimes based on the direction of growth and inflation relative to expectations. Each indicator in this guide feeds one axis; policy rates, yields, the dollar and gold show how markets are pricing the regime.

Exhibit 55: Growth x inflation regimes

Regime	Typical indicator signature	Asset tendencies
Goldilocks (growth up, inflation down)	PMIs rising above 50, payrolls above breakeven, permits rising, core CPI/PCE decelerating, VIX low	Equities (esp. Nasdaq, SOX), credit; curve stable; USD mixed
Overheating (growth up, inflation up)	ISM New Orders and Prices both high, wages and ECI hot, quits high, capacity utilisation above 80%, oil rising	Commodities, value, energy, financials; short duration; Fed hawkish; bear flattening
Stagflation (growth down, inflation up)	PMIs < 50 with Prices > 60, oil spike, sentiment collapsing, rising inflation expectations	Cash, commodities, gold, short-dated TIPS; equities and long bonds both suffer
Disinflationary slowdown / recession (growth down, inflation down)	Payrolls below breakeven, Sahm rule rising, claims and layoffs up, hires and quits falling, permits falling, VIX elevated, oil falling	Long Treasuries (bull steepening), quality and defensives, USD vs high-beta FX; Russell 2000 underperforms until cuts are priced

Source: compiled for this guide.

A monthly workflow (with Taipei times)

Taipei times below apply while the US is on standard time (early November to mid-March); during US daylight saving time, subtract one hour.

Exhibit 56: A monthly data workflow

When (approx.)	Release	Taipei time	What to update
1st business day	ISM Manufacturing; S&P Global final mfg PMI	23:00 / 22:45	Industrial cycle, prices pipeline
3rd business day	ISM Services; S&P Global final services PMI	23:00 / 22:45	Services momentum and prices
Tuesday after month start	JOLTS (for two months earlier)	23:00	Labour flows, quits
Wednesday before NFP	ADP National Employment Report	21:15	Private payroll preview
First Friday	Employment Situation (NFP, unemployment, AHE)	21:30	Labour market, Fed mandate
Early month	Trade balance; World Bank Pink Sheet	21:30	Net exports, commodity prices
~8th business day	Monthly Treasury Statement	03:00 (next day)	Deficit, tariff revenue
~2nd Friday	Michigan preliminary	23:00	Household inflation expectations
~10th-15th	CPI, then PPI	21:30	Inflation trend; implied core PCE
~15th-17th	Industrial production & capacity utilisation	22:15	Output, slack
~17th-19th	Housing starts & permits	21:30	Rate-sensitive cycle signal
~3rd-4th week	S&P Global flash PMIs; existing & new home sales; Michigan final	22:45 / 23:00	Earliest growth read; housing
Last days of month	PCE, personal income & spending; GDP (quarterly); ECI (quarterly)	21:30	Fed target measure, consumption, labour costs

When (approx.)	Release	Taipei time	What to update
8 times a year	FOMC decision; SEP and dot plot quarterly; minutes 3 weeks later	03:00 (next day)	Policy path vs FedWatch pricing
Weekly	Jobless claims (Thu); EIA petroleum (Wed) and gas storage (Thu)	21:30; 23:30	Labour stress, energy balances
Quarterly / semi-annual	Current account; IMF WEO (Apr/Oct); World Bank GEP (Jan/Jun)	21:30; scheduled	External balance, global context
Continuous	Treasury yields, DXY, gold, oil, equity indices, VIX, CME futures	Nearly 24h	Real-time regime check

Source: agency release calendars.

Principles worth internalising

- **Surprises and revisions drive markets**, not levels. Track consensus, FedWatch pricing and nowcasts.
- **Triangulate:** one survey or one month is noise; agreement across soft data, hard data and markets is signal.
- **Know the Fed's reaction function:** the same data can be bullish or bearish for equities depending on whether inflation or growth is the binding constraint.
- **Follow the chain:** data change Fed expectations (CME futures), which move the 2-year yield, the dollar and equity multiples; fiscal supply and term premium drive the long end.
- **Watch structural shifts:** lower immigration (breakeven payrolls), AI capex (investment, IP, SOX, power demand), tariffs (core goods, ISM prices, trade and fiscal data), large deficits (term premium) and reserve diversification (gold, COFER) change how indicators should be interpreted.
- **Use vintage data** (ALFRED) when backtesting - revised data overstate how predictable things were.

Appendix: free data toolkit

Exhibit 57: Free data toolkit

Tool	Access	Best for
FRED (fred.stlouisfed.org)	Web, Excel add-in, API (free key); Python: fredapi or pandas-datareader	One-stop for most US series in this guide
ALFRED (alfred.stlouisfed.org)	Same interface	Historical vintages / first releases
BLS API (api.bls.gov)	Free registration key	CPI, employment, wages, JOLTS detail
BEA API (apps.bea.gov/API)	Free registration key	GDP, PCE, international transactions
Census (census.gov/data; api.census.gov)	Free key	Housing, trade, retail sales
Federal Reserve DDP (federalreserve.gov/datadownload)	CSV downloads	Industrial production, H.15 rates, H.10 dollar
Treasury (fiscaldata.treasury.gov; home.treasury.gov)	API, CSV	Budget, debt, yields, auctions
CME Group (cmegroup.com)	Web; free login for some tools	FedWatch, futures settlements, volume, open interest
EIA Open Data (eia.gov/opendata)	Free key	Oil and gas prices, inventories, production
IMF (data.imf.org) and World Bank (data.worldbank.org)	Free, SDMX / REST APIs	Global forecasts, cross-country and long-run data
Release calendars	bls.gov/schedule, bea.gov/news/schedule, federalreserve.gov (FOMC calendar), ismworld.org, pmi.spglobal.com; Investing.com/ForexFactory for consensus (can display Taipei time)	Planning and consensus

Tool	Access	Best for
Nowcasts	Atlanta Fed GDPNow, NY Fed Staff Nowcast, Cleveland Fed Inflation Nowcasting	Expectations between releases

Source: publisher websites, compiled for this guide (access terms as of November 2026).

Minimal Python example: `from fredapi import Fred; f = Fred(api_key='YOUR_KEY'); core = f.get_series('PCEPILFE'); print(core.pct_change(12).tail())`

Disclaimer: This guide is an educational reference prepared in November 2026. It is not investment advice. Statistical agencies, central banks and index providers revise methodologies, weights, schedules and access terms; figures such as component weights, panel sizes, thresholds, shares and breakeven estimates are approximate. Confirm current details on each publisher's website.